

Immigration and Credit in the United States

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Abstract

We study immigrant assimilation into U.S. consumer credit markets, identifying individual immigrants in credit reporting data from the timing of Social Security Number assignment. Immigrants arrive without U.S. credit histories yet are positively selected: credit scores are approximately 25 points higher than non-immigrants' throughout their thirties. Despite greater creditworthiness, by age 37, immigrants are 16 and 9 percentage points less likely to have had an auto loan and a mortgage, respectively. Same-age immigrants arriving one year later-in-life have lower auto loan and mortgage use for over a decade. These patterns suggest history dependence: time in the U.S. shapes credit access.

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1. INTRODUCTION

One in seven U.S. residents was born abroad, and essentially all began American life without a credit file. Immigrants arrive in the United States with their credentials and work experience, which help them navigate U.S. labor markets. By contrast, credit histories do not cross borders: lenders cannot see how immigrants repaid their debts in their home countries, which may limit their access to the U.S. financial system. This information friction matters because credit underpins the purchases through which American households build wealth and integrate into society.

We provide large-scale evidence on how immigrants assimilate into the U.S. consumer credit system. To identify legal immigrants and their arrival timing at the individual level, we use variation in the lifecycle timing of Social Security Number (SSN) assignment: people assigned SSNs as adults (at age 21+) are predominantly immigrants (Klopfer and Miller, 2024, Bernstein et al., 2025).¹ Our dataset is a 10% representative sample of U.S. consumer credit reporting data from TransUnion spanning 2000 to 2024. We make two main same-age-and-geography comparisons: first, between non-immigrants and immigrants, and second, between immigrants who arrived in the U.S. at different ages. Our findings provide new facts about immigration and credit in the United States, as well as evidence on mechanisms.

We begin with selection, evaluating credit scores, which are a key input into credit decisions and are often interpreted as a summary statistic for financial well-being (Gibbs et al., 2025, Bakker et al., 2026). Once immigrants enter the credit system, their average credit score is 27 points higher by age 30 than same-age non-immigrants in the same geographic area. This average difference persists as consumers age: the distribution of credit scores by age 40 is systematically higher for immigrants than for non-immigrants (Figure 1). This positive selection on credit quality is more pronounced among those who immigrate when they are older, and is not specific to credit scores – immigrants also have lower delinquency rates and lower credit utilization.² Moreover, even conditional on credit score at age 30, immigrants are of higher credit quality: 1.5 percentage points (5.5% of the mean) *less* likely than non-immigrants to become delinquent within seven years.

¹We validate this classification against ACS data, as described in Section 2.2 and Appendix A.1.

²This positive selection is not driven by a predominance of high-skilled visa holders. Legal immigrants to the United States have a broad range of immigration statuses, including employment-based, immediate relative, and family-sponsored. Most legal immigrants to the United States are *not* H-1B visa holders (a type of visas for workers in specialty occupations, often requiring higher education). The Office of Homeland Security Statistics Immigration Statistics (2024) show that from 2004 to 2023, only 16% of legal immigrant temporary workers admitted were on H-1B visas, and only 9% of legal new arrivals awarded lawful permanent residence (green cards) for employment are professionals with advanced degrees or exceptional ability. Appendix Figure A1 shows how most immigrants have fewer years of education and lower income than non-immigrants.

Despite having higher credit quality than non-immigrants, immigrants access credit later, and for asset-linked credit, many have not caught up by the end of our sample (i.e., by age 37). At this point, immigrants are 16 percentage points (pp, 21% of the mean) less likely to have ever had an auto loan and 9 pp (20%) less likely to have had a mortgage than same-age non-immigrants in the same geographic area. Timing is central to this pattern: each year of later arrival in an immigrant’s twenties is associated with 1.6 and 1.8 pp lower auto loan and mortgage use, respectively, by age 37. Even as credit card access effectively converges, card limits – an intensive margin of credit – lag for up to a decade, which also appears to reflect immigration timing. These credit access gaps are not fully driven by credit scores: they persist even within credit score bins.

We argue that our results reflect *history dependence*: credit access depends on the length of a consumer’s time in the U.S., beyond what their credit score captures. Whether a consumer can access a credit card depends primarily on their credit score, whereas credit card limits, access to auto loans, and especially mortgages also depend on broader information about a consumer (Agarwal et al., 2018, Blattner et al., 2022, De Giorgi et al., 2023, Grunewald et al., 2026, Kovrijnykh et al., 2026, Raveendranathan et al., 2026) that accumulates with time in the country. History dependence makes two predictions. First, among same-age immigrants, credit access should decline with age at arrival, since later arrivals have less U.S. history. Second, this decline should be steeper for auto loans and mortgages than for holding a credit card. Both predictions hold: each year of later arrival is associated with markedly lower auto loan and mortgage access but only slightly lower likelihood of holding a credit card.

To trace how credit access evolves around immigration’s timing, we use a paired cohort design that compares immigrant cohorts born in the same year but who arrived one year apart. This comparison, which excludes non-immigrants, captures the relationship between one additional year in the U.S. and long-run credit access. One year of later arrival is associated with auto loan and mortgage gaps that do not close, in contrast to credit card differences, which disappear. Thirteen years after immigration, the later cohort remains 1.1 pp less likely to have had an auto loan and 1.4 pp less likely to have had a mortgage. Moreover, most of the auto and mortgage gap emerges many years after arrival – when these products are typically first taken up – rather than mechanically in the year of arrival itself.

To assess whether credit access gaps between immigrants and non-immigrants merely reflect differences in income or observable socioeconomic characteristics, we complement the credit reporting data with public data from the American Community Survey (ACS), using 5-year waves covering

2005 to 2023. We show that immigrants arriving in their twenties are less likely to have a mortgage or own a home than non-immigrants in the same birth cohort, even after controlling for geography and detailed demographic characteristics. Immigrants have substantially lower household income, and lower with age at arrival. Yet, the mortgage and homeownership gaps and their decline with age at arrival persist when we add demographic and income controls, suggesting that immigrants' lower credit access does not only reflect their lower incomes.

We evaluate alternative explanations for our results and find that none fully accounts for them. Emigration cannot: our results are robust to conditioning on consumers still present at the end of our data, and later in life arrivals are less likely to emigrate yet have lower access, the opposite of what an emigration story predicts. Demand differences exist, as immigrants make fewer auto loan or mortgage inquiries and hold more negative attitudes toward auto debt, but demand does not appear to vary with age at arrival, so it cannot explain the decline in access with later arrival. Supply-side evidence points the same way as history dependence: immigrants are more likely to be rejected for auto loans or mortgages than non-immigrants, controlling for credit score. Cultural factors do not explain our results: the timing of immigration matters within region of origin, within country of origin, and within terciles of home-country homeownership rates. All of these factors may contribute to credit differences between immigrants and non-immigrants, but none of them explains why access declines with age at arrival.

Our paper provides a fresh perspective on the assimilation of immigrants into the U.S. economy. There is a large literature on immigrant assimilation, which has characterized immigrants' labor market, cultural, and social integration during different immigration waves (e.g., Borjas, 1985, Abramitzky et al., 2014, 2020, 2021, Lubotsky, 2007, Bleakley and Chin, 2010, Bailey et al., 2026, Akee et al., 2026, Kerr et al., 2026). Our evaluation of immigrants' entry into U.S. consumer credit markets not only offers a distinct venue – credit markets versus labor markets – but also suggests that the inability to port credit histories across national boundaries may slow assimilation into the credit system and delay home and vehicle ownership for immigrants.

Our findings also inform how immigrants are selected, a major theme in the immigration literature. As reviewed in Abramitzky and Boustan (2017), immigrants to the U.S. have swung between being positively selected and negatively selected, depending on the measure of selection, the time period, and the country of immigration. We contribute to this literature by characterizing immigrant credit quality using detailed consumer credit reporting data. Creditworthiness is an important characteristic that differs from the labor market outcomes primarily studied in the im-

migration literature. Income and other labor market characteristics (e.g., education, employment) are not inputs to credit scores (Gibbs et al., 2025) and credit scores have little correlation with income (Berman et al., 2026). Indeed, Chatterjee et al. (2023) consider a credit score to be the market’s assessment of a person’s unobservable type, which they interpret as patience. Our finding that immigrants are highly and positively selected relative to non-immigrants highlights the latent quality of immigrants and is consistent with immigrants’ potential to contribute to the American economy (Bernstein et al., 2025).

Our analysis also relates to recent work on the dynamics of credit and geographic mobility. Much of this work studies how consumer outcomes track moves within the United States (Brown et al., 2019, Keys et al., 2023, Bakker et al., 2026). Immigrants enter the credit system as adults, so the age of their file is decoupled from their own age, and distinct from other reasons for later arrival in the credit system, such as the role of childhood experiences or parental histories (e.g., Bach et al., 2023, Bakker et al., 2026). By studying immigrants – an important consumer segment – we show that the timing of credit market entry matters for credit use later in life.

Within household finance, our contribution is to systematically document immigrants’ assimilation into the credit system and to highlight history dependence as a key contributing mechanism. Existing research on immigration has largely evaluated whether cultural factors shape financial behavior and outcomes (e.g., Carroll et al., 1994, Osili and Paulson, 2004, Haliassos et al., 2017, Fuchs-Schündeln et al., 2020, Zillessen, 2026, Gorback and Schubert, 2026), as reviewed in Gomes et al. (2021). One exception is Dobbie et al. (2021), who show that a subprime lender’s decisions in the United Kingdom are biased against immigrants. To our knowledge, history dependence has not been studied empirically, but it is consistent with recent models of credit access outside the immigration context (e.g., Kovrijnykh et al., 2026, Raveendranathan et al., 2026). In contrast to cultural explanations, which emphasize differences in preferences and beliefs, the decline in access with age at arrival persists irrespective of country of origin.

The paper proceeds as follows. Section 2 explains our data, how we classify immigrants, institutional details of lending to immigrants, and our empirical strategy. Section 3 shows our results. Section 4 contains the methodology and results for the paired cohorts analysis. Section 5 evaluates the mechanisms behind our results and Section 6 concludes.

2. DATA AND EMPIRICAL DESIGN

In this section, we describe the data that we use and explain how we classify immigrants and their year of immigration. Subsection 2.3 provides context on lending to immigrants in the United States, while subsection 2.4 explains our empirical methodology.

2.1 DATA

2.1.1 CONSUMER CREDIT REPORTING DATA

We use consumer credit reporting data from the University of Chicago Booth TransUnion Consumer Credit Panel, “BTCCP” (TransUnion, 2026). The BTCCP is an anonymous, representative sample of U.S. consumer credit reporting data provided by TransUnion that contains monthly information from July 2000 to December 2024. To build this sample, TransUnion starts with a 10% representative sample of consumers who had a credit report in July 2000 and then 10% of new consumers entering the main TransUnion data each month are added to the BTCCP to maintain its representativeness. We follow the best practices in Gibbs et al. (2025) for using such data.

The BTCCP is a collection of datasets, linkable to one another via a consumer identifier. We primarily construct our outcomes from the BTCCP’s monthly tradeline-level dataset, which contains information (i.e., outstanding balance, delinquency status), including the date each account was opened, on each credit account held by a consumer. This tradeline dataset includes accounts opened or closed prior to July 2000. We also use the BTCCP’s consumer-level header dataset, which contains each consumer’s birth date and the date a consumer first has a credit report (including if before July 2000). By combining the consumer’s birth date with the account opening date in tradeline data, we compute the consumer’s age at account opening.³ From the BTCCP’s monthly consumer-level aggregated dataset we use a consumer’s commercially-available credit score, their ZIP code and state, and months since the last credit inquiry. We use these monthly datasets to construct an annual panel of credit characteristics as of July of each year, and use monthly data to calculate control variables for consumers’ first credit score and based on their geographic location. The final dataset is the inquiries file, where each credit inquiry between 2009 and 2019 is listed with the date and type of credit applied for but is only available for a random sample of consumers.

³We follow standard practice in removing tradelines not been updated in the last twelve months (Gibbs et al., 2025). We drop tradelines opened before the age of 18 to produce more plausible estimates of credit usage at age 18, as these cases are before a consumer can take out a credit agreement by themselves, and are likely to either be a data error or a credit product taken out by their parent. Such restrictions do not affect our results.

2.1.2 AMERICAN COMMUNITY SURVEY (ACS)

We use public Census data from the 5-year American Community Survey (ACS), accessed via the Integrated Public Use Microdataset Series (Ruggles et al., 2025). These anonymous data contain birth years, years of immigration, countries of birth, geographic locations, and detailed socio-economic information. We use the 2010 wave as a benchmark to evaluate our immigrant classification. We also use 5-year repeated cross-sectional waves that cover the years from 2005 to 2023 to evaluate the mechanisms behind our results. We merge international homeownership rate data from Gorback and Schubert (2026).

2.1.3 SURVEY OF CONSUMER FINANCES (SCF)

We use public data from the nationally-representative cross-sectional Survey of Consumer Finances, SCF, (Board of Governors of the Federal Reserve System, 2023) to help evaluate the mechanisms behind our results. We use the 2022 wave of the SCF as this is the only wave that includes information on whether a consumer is an immigrant and the number of years since immigration. The SCF contains detailed socio-economic information, however, does not contain geographic information or credit scores. See Appendix A.2. for additional details about the SCF.

2.2 IMMIGRANT CLASSIFICATION

In our credit reporting data, we classify whether a consumer is an immigrant and their age at immigration following the “sequential SSN assignment” procedure used in recent research on U.S. immigration (Yonker, 2017, Doran et al., 2022, Bernstein et al., 2025, Klopfer and Miller, 2024).⁴ SSNs are nine-digit numbers in use since 1936 that are unique to a consumer and are assigned by the Social Security Administration (SSA). Up until mid-2011, the SSA sequentially assigned SSNs using the same first five digits before moving onto another five-digit block, with the timing of these allocations being public. The first three digits denote an “area number,” a geographic region within a state, the fourth and fifth digits are the “group number,” assigned sequentially in blocks within that area, and the last four digits are the order of processing, which is quasi-randomly assigned (see Puckett, 2009 for a comprehensive history of SSN assignment). As established in prior research,

⁴The closest prior data to ours is Federal Reserve (2007) report to Congress, which evaluated traditional credit scoring models in a 2003 sample of 300,000 credit records that included immigrants. Our dataset offers a substantial advancement over this early work because it is more recent, more comprehensive, more granular, and has information on immigration age. Because of these features and because our data cover 25 years, we are able to speak to the question of assimilation into U.S. credit markets by evaluating the dynamics of immigrant credit.

consumers assigned an SSN at age 21 or later in life are generally immigrants (Klopfer and Miller, 2024, Bernstein et al., 2025).

TransUnion observes SSNs, however, the BTCCP that we have access to is anonymized without SSNs. We therefore developed a procedure to classify immigrants in our data without directly accessing SSNs. To estimate the year of SSN assignment, *SSN Year*, we construct a lookup table using public information that maps the first five digits of the SSN into the first year of SSN assignment, *SSN Year*, for all SSNs assigned before 2012. We sent TransUnion a list of anonymous consumer identifiers in the BTCCP, together with this lookup table. TransUnion then matched the consumer list and the lookup table to their underlying data, which includes consumers' SSNs, returning a dataset with the year of SSN assignment, an indicator for whether a consumer had any SSN in their data, and the BTCCP consumer identifier. This procedure guarantees that we neither observe nor can infer SSNs for any consumers in the BTCCP. We follow Bernstein et al. (2025) in classifying consumers as an immigrant if their age at SSN assignment, *SSN Age*, is greater than or equal to 21, where $SSN\ Age = SSN\ Year - Birth\ Year$.

We apply two sample restrictions. First, we restrict to consumers who have an SSN in TransUnion and whose birth years are between 1975 and 1987. Consumers with these birth years are mostly expected to enter the credit system after 2000 (and in our data we observe their accounts that were opened before 2000) with their SSNs assigned by 2011. For example, from 2000 to 2024 (the BTCCP coverage window), consumers born in 1975 are observed from ages 25 to 49, while consumers born in 1987 are observed from ages 18 to 37. This part of the lifecycle – early adulthood to middle age – is when we expect credit access to be most economically important. This follows similar approaches used in related papers (Bach et al., 2023, Bakker et al., 2026). Restricting to birth years 1975 to 1987 also corresponds to the birth years in our data where the estimated immigration shares closely resemble those in the ACS, as shown in Panel A of Appendix Figure A2. For some analyses that rely on outcomes constructed from the BTCCP's consumer-level datasets, we further restrict to birth years 1982 to 1987 to ensure that consumers enter the credit system after 1999. Second, we restrict our attention to consumers with $0 \leq SSN\ Age < 30$, which focuses our analysis on immigrants arriving in the United States at some point in their twenties. This ensures that, for all consumers, credit outcomes in a consumer's thirties occur *after* everyone in our sample has immigrated and we can examine longer-term credit outcomes observed by 2025.

After applying these restrictions, we observe 6,299,287 consumers, 6.1% (384,396) of whom we classify as immigrants (i.e., SSN Ages 21 through 29). This is consistent with the ACS data

estimate of 6.6% (after applying the same birth year and immigration age restrictions), and the distribution of immigration ages is consistent between our data and the ACS (Appendix Figure A3). Although some older consumers are assigned SSNs as adults, for the birth years that we study, non-immigrants are largely assigned an SSN at birth or before turning 18, as demonstrated in administrative Social Security Administration data by Klopfer and Miller (2024).⁵ Reassuringly, the consumers we classify as immigrants are more geographically mobile (Appendix Figure A5).

To help evaluate the external validity of our data restrictions, we examine the 2010 5-year ACS data. We find a similar distribution of income and education levels in the ACS with and without imposing our sample restrictions (Appendix Figure A1). Compared to non-immigrants, immigrants typically have lower incomes and are generally less educated. Immigrants are more likely than non-immigrants to be educated at the doctoral level, however, this represents only 2.1 percent of immigrants. Additional details on data construction and validation are in Appendix A.1.

Our classification of immigrants excludes certain groups. First, it does not include illegal immigrants (who lack SSNs), and who are less likely to use the formal credit system for demand and supply reasons. Second, we do not observe immigrants or non-immigrants who remain entirely outside the formal credit system, because we study formal credit only, not informal credit. Not observing informal credit use is a minor concern since our main outcomes — mortgages and auto finance — involve assets that are hard to finance informally. Third, we do not separately study immigrants who arrive before adulthood or at older ages. Childhood immigrants cannot be accurately classified in our data as many non-immigrants are assigned SSNs at different ages of childhood (Klopfer and Miller, 2024). We focus on younger age immigrants as this is the stage of the lifecycle where credit access is most important. Fourth, we do not include legal immigrants without SSNs, such as those who only have Individual Taxpayer Identification Numbers (ITINs) or Enumeration at Entry numbers (EAE), as we do not know when these are assigned. ITINs and EAEs are typically used before a consumer receives an SSN. Consumers who temporarily had ITINs (or EAEs) before later being assigned SSNs would be in our data.

⁵If we instead classified immigrants as those with SSN ages greater than 18 or greater than 19 the comparison is unaffected, as the non-immigrant baseline is almost unchanged and only 109,207 consumers have SSN Ages of 19 or 20 – see Appendix Table A1, Panel B. Appendix Figure A4 shows the distribution of SSN Ages by birth years in our data, which mirrors that in Klopfer and Miller (2024).

2.3 INSTITUTIONAL DETAILS ON LENDING TO IMMIGRANTS

Lenders in the United States may consider an applicant’s immigration status in their lending decisions because immigration status is *not* a protected characteristic under the Equal Credit Opportunity Act (ECOA). This contrasts with ECOA-protected characteristics such as gender, national origin, and race. Immigration status is observed in auto loan and mortgage lending applications as a by-product of requesting documents (e.g., passports) to verify the identity of an applicant. For credit card applications, a lender may only know whether an applicant has an ITIN or an SSN. As detailed by the Consumer Financial Protection Bureau (2023), until 2026 lenders have been constrained in how much they can rely on immigration status in their underwriting as they need to avoid discriminating on ECOA-protected characteristics, however, visa status may limit which types of mortgages an immigrant is eligible for. Since 2025, government policy has shifted with the announcement of new measures that are expected to directly restrict mortgage credit to immigrants (U.S. Department of Housing and Urban Development, 2025, Consumer Financial Protection Bureau, 2026). Financing auto loans to immigrants appears to be becoming harder since 2025 with one auto lender (Lendbuzz) warning of increased risks of lending to immigrants in its 2025 filing ahead of its Initial Public Offering. Such changes in the environment are too recent to affect our results.

Lenders in the United States use credit scores and other information in their lending decisions. Information frictions may affect lenders’ decisions and impact immigrants’ access to credit in several ways. First, immigrants initially lack a U.S. credit report and a credit score. Lenders may therefore have a prior that, given no information, immigrants are high credit risk, as in the model of Pagano and Jappelli (1993). Second, after entering the U.S. credit system, immigrants have a shorter length of credit history than most non-immigrants of the same age, who often enter the credit system at age 18 or in their early twenties. The length of credit history positively affects credit scores (VantageScore, 2019, FICO, 2025). Separately, the length of credit history may also be a direct input in lenders’ underwriting via overlays applied in combination with scores (e.g., applicants may need at least two years of credit history irrespective of score) or as an input into proprietary scores that are created by lenders (Gibbs et al., 2025).

The length of credit history can also have indirect impacts on credit access, as, in the U.S. credit system, consumers need to take out credit to access credit. Less time in the credit system means less time to grow the number of credit accounts on a consumer’s credit report. This can impede

credit access, as it makes immigrants more likely to have “thin files”, where they hold only a few credit accounts, which results in noisier assessments of credit risk (Blattner and Nelson, 2024). A shorter credit history can also mean less time for credit card limits to grow. This can reduce credit access because lenders increase card limits over time and they also are influenced by the existing limits assigned to a consumer by competing lenders (De Giorgi et al., 2023, Kovrijnykh et al., 2026, Raveendranathan et al., 2026).⁶

Cultural barriers may also prevent immigrants from accessing credit, such as differences in familiarity and willingness to use or repay debt. Language barriers or a lack of financial literacy may mean that immigrants may not understand the U.S. credit system, such as how one needs to take out and use credit to build and improve their credit score.

2.4 EMPIRICAL METHODOLOGY

We use two main empirical specifications to analyze outcomes in our credit reporting data. For both of these specifications, we collapse the data to one observation per consumer i estimating OLS cross-sectional regressions. Our first main specification is Equation (1):

$$Y_i = \beta_1 \cdot \mathbb{I}\{Immigrant\}_i + \mu_{b(i)} + \eta_{g(i)} + \epsilon_i, \quad (1)$$

In this specification, β_1 is the coefficient of interest, estimating the average difference in an outcome, Y_i , between immigrants and non-immigrants. This is the coefficient on an indicator variable, $\mathbb{I}\{Immigrant\}_i$, which takes a value of one for *immigrants* – consumers with SSNs assigned at ages 21 through 29 – and takes a value of zero if their age of SSN assignment is less than 21, who we classify as *non-immigrants*. This regression also includes birth year fixed effects ($\mu_{b(i)}$), as well as geographic fixed effects ($\eta_{g(i)}$) for the first five-digit ZIP code (First ZIP5) associated with a consumer in our monthly data, to capture unobservable heterogeneity in local market conditions. Including such geographic controls is important as geographies vary in their economic opportunities and access to finance, and immigrants may select into their location partially due to such factors.

Our second main specification is Equation (2):

$$Y_i = \beta_1 \cdot \mathbb{I}\{Immigrant\}_i + \beta_2 \cdot SSN\ Age_i + \mu_{b(i)} + \eta_{g(i)} + \epsilon_i, \quad (2)$$

In this specification, we include a linear term for the age of SSN assignment ($SSN\ Age_i$) to

⁶Other factors could also lead lenders to restrict credit to immigrants. For example, in the United Kingdom, Dobbie et al. (2021) show how one subprime consumer lender’s decisions are biased against immigrants due to loan examiner incentives focused on short-run outcomes, rather than long-run profits.

allow for heterogeneity in immigrant outcomes. To aid interpretation, we pool all non-immigrant SSN Ages (0 through 20) into one category. The β_2 coefficient reflects the marginal difference in average outcomes associated with being assigned an SSN one year later in life. In both of these specifications, we cluster standard errors by birth year $\times$ the first two digits of a consumer’s first observed ZIP code (a coarser unit of geography than ZIP5 that corresponds to state or subregion).

In the Appendix, we show that our results are robust to adding fixed effects for additional geographic controls: the most recent five-digit ZIP observed for a consumer in our data (*Last ZIP5*), the five-digit ZIP observed for a consumer for the most months in our data (*Longest ZIP5*), which is a measure of their most permanent location, and the number of unique five-digit ZIP observed for a consumer (*Number ZIP5*) which is a measure of their geographic mobility. All of these geographic controls are imperfect because consumers enter and exit our data at different times, partially based on their credit access. In the Appendix, we also show that our results are robust to restricting to consumers who have not emigrated, which we proxy by conditioning on the subsample of consumers who are still observed in the data in 2024.⁷

3. RESULTS

This section describes our main results, with an alternative empirical design presented later in Section 4. Table 1 presents sample means of variables for immigrants (*SSN Age* 21+) versus non-immigrants (*SSN Age* < 21). These show clear differences in the nature and timing of immigrant versus non-immigrant credit use. Immigrants typically enter the U.S. credit system later than non-immigrants, have higher credit scores, and are less likely to hold auto loans or mortgages.

3.1 CREDIT MARKET ENTRY

We demonstrate that the age at which a consumer receives an SSN (*SSN Age*) drives the timing of credit market entry. This provides an internal consistency check on *SSN Age* as a proxy for immigration timing. Table 2 presents the results from estimating our regression specifications (equations (1) and (2)). Columns 1 and 3 show that immigrants are 5.5 years older than non-immigrants when they receive their first credit report, and 5.2 years older when they receive their first credit product, after controlling for birth year and first ZIP5 via fixed effects. In columns

⁷Because our main dataset (without the emigration restriction) is a longitudinal panel, we can follow cohorts over time. In the immigration literature, this is known as a “complete” dataset (i.e., we can consistently follow cohorts from their age of SSN). This structure helps limit the sampling biases or compositional changes for estimating immigrants’ earnings in repeated cross-sections or household panels with attrition outlined in Dustmann and Görlach (2015).

2 and 4, we add SSN Age to the specification finding a baseline delay for immigrants relative to non-immigrants is 2.3 and 2.1 years, for first credit report and product, respectively. For each additional year of later in life SSN assignment, the first credit report (product) is delayed by an additional 0.75 (0.73) years, economically significant relative to the average age of 20.2 (21.6) that non-immigrants receive their first credit report (product). We interpret this delay as primarily mechanical: immigrants could not obtain credit in America prior to immigrating.⁸ These results are also robust to adding fixed effects for the consumer’s last ZIP5 and longest ZIP5, and number of unique ZIP5s (Appendix Table A2). Panel A of Appendix Figure A6 shows an essentially linear increase in the average age at first credit product for each additional year of SSN Age.

3.2 CREDITWORTHINESS

A core aim of the immigration literature is to understand how immigrants assimilate into the U.S. economy (e.g., Borjas, 1985). Our data enable us to measure how a consumer’s creditworthiness and credit access evolves over their lifecycle via two comparisons: comparing non-immigrants to immigrants, and comparing the outcomes of immigrants assigned SSNs at different ages. In this section, we provide evidence on selection of immigrants, and then in the two following sections we study how immigrants assimilate into U.S. credit markets.

3.2.1 FIRST CREDIT SCORE

To complement the cross sectional regressions, in Figure 2 and later analyses that follow, we visually summarize our data for each combination of *SSN Age* and *Age* observed in our panel data, grouping consumers assigned an SSN at age 18 or earlier into a single benchmark category of non-immigrants with black lines. Other SSN Ages contain immigrants and have shades of blue lines, with lighter shades for consumers assigned SSN Ages later in life. This figure, and subsequent analogous figures, show SSN Ages 19 and 20 separately, to illustrate how classifying these consumers, contain a mixture of immigrants and non-immigrants, would not impact our conclusions (consistent with Bernstein et al., 2025, whose results are robust to classifying immigrants based on SSN Ages 19+, 20+, or 21+).

We examine the timing of a consumer’s first credit score, as measured by a commercially-available credit score. Panel A of Figure 2 displays the unconditional likelihood of having a credit

⁸One might expect that each additional year of SSN Age would mechanically delay the consumer’s entry by one year. However, we would expect a distribution of immigrants’ years of entry and not all non-immigrants have entered the credit system at each age (Panels A and B of Appendix Figure A7) which makes the mechanical effect below one.

score for each SSN Age cohort at different ages. At age 18, 14% of the SSN Age 18 cohort of non-immigrants had received their first credit score; by age 22, 84% of these consumers have a score. The consistent rightward shift in these lifecycle curves as SSN Age increases indicates that consumers with later in life SSN Ages also receive a score later in life. In the years prior to the SSN Age (where $Age = SSNAge$ is indicated by a circle on each lifecycle curve), the likelihood of having a credit score is very low, and this is followed by a sharp jump in the percentage of scored consumers in the year following immigration. A small share of consumers have a credit score slightly before they receive an SSN because it is possible to access credit without an SSN (e.g., some consumers will have an ITIN before receiving an SSN).

Shortly after their immigration year, immigrants (i.e., those with SSN Ages 21 to 29), quickly converge to the SSN Age 18 non-immigrant cohort (the black line in the figure) in their likelihood of having a credit score. The convergence is quicker for immigrants assigned SSNs later in life. Moreover, immigration cohorts arriving later in life have a higher likelihood of having a credit score in their year of immigration. For example, only 8% of 21 year old immigrants (SSN Age of 21) have a credit score by age 21 whereas 22% of 29-year-old immigrants have a credit score by age 29.

3.2.2 MEAN CREDIT SCORES

It is important to study credit scores at different ages because a consumer’s borrowing needs and propensity to repay debt can change over the course of their lifecycle (see Chatterjee et al., 2023, for a theory of credit scoring). Panel B of Figure 2 plots the mean credit score for each SSN Age cohort at different ages, *conditional* on having a credit score at that age. This plot shows that immigrant SSN Age cohorts (SSN Age 21+) start out with a higher mean credit score than the (non-immigrant) SSN Age 18 cohort has at the same age. Immigrant cohorts continue to have substantially higher average credit scores throughout their thirties. Although the high average credit score in the year of immigration may partly reflect selection (i.e., immigrants who are scored are of higher credit quality), immigrant cohorts have higher credit scores than the SSN Age 18 cohort even once a similar fraction of both the immigrant cohorts and the SSN Age 18 cohort are scored. This suggests that immigrants who are scored are more creditworthy than their scored non-immigrant counterparts of the same age, on average. Overall, these dynamics paint a somewhat unexpected picture: Even new immigrants with a credit score are *observably* more creditworthy.

To provide a more formal characterization of these lifecycle patterns, in Panel A of Table 3 we report estimates from equations (1) and (2) for how average credit scores by ages 30 and 40 depend

on immigration status and SSN Age. These regressions control for first geographic location (ZIP5) and birth year, neither of which are inputs to credit scores (Gibbs et al., 2025). Consistent with the descriptive lifecycle plots, column 1 shows that immigrants have average credit scores by age 30 that are 26.7 points higher than non-immigrants. In column 2, we allow for heterogeneity by SSN Age, and find that the baseline difference in credit score by age 30 between immigrants and non-immigrants is 17.2 points, and increases by an additional 2.3 points per year of SSN Age.

The difference in average credit scores between immigrant and non-immigrants persists as consumers age. Column 3 of Panel A of Table 3 shows that immigrants’ average credit score is 24.6 points higher by age 40. Column 4 accounts for heterogeneity by SSN Age, showing that the baseline difference is 14.2 points and increases by an additional 2.4 points per year for consumers who immigrate later in life. These higher average credit scores occur *despite* these consumers having a shorter credit history — the length of credit history is an important input into credit scores, accounting for approximately 15% of FICO and 20% of VantageScore models (VantageScore, 2019, FICO, 2025). Because immigration status is not an input into credit scores, immigrants’ higher scores reflect differences in credit quality between immigrants and non-immigrants.

3.2.3 DISTRIBUTION OF CREDIT SCORES

The mean credit score may mask differences across the distribution. Once scored, immigrants have a higher quality distribution of credit scores than non-immigrants of the same age. Panel B of Table 3 estimates the probability of attaining a prime-or-higher score (credit score > 660), assigning consumers with lower or no scores a value of zero. Immigrants are 4.6 pp more likely to attain a prime-or-higher score by age 30 and 11.2 pp more likely by age 40, a large difference relative to a 49.3% non-immigrant mean at age 40. At age 30, each additional year of SSN Age is associated with a 0.5 pp lower probability of a prime-or-higher score, primarily because later-arriving immigrants are less likely to have been scored. By age 40, effectively all consumers are scored, and each additional year of SSN Age is associated with an approximately 1 pp higher probability. These results are robust to additional geographic fixed effects (Appendix Table A3). Panels C and D of Figure 2 illustrate the role of unscored consumers. Panel C includes them in the denominator, while Panel D conditions on having a score. During their twenties, later-arriving cohorts’ lower probability of being scored outweighs their higher scores conditional on being scored. During their thirties, the latter difference (higher scores) dominates. Conditional on having a score, immigrants are substantially more likely than non-immigrants to have prime-or-higher credit.

The longer a consumer’s credit history, the more time there is for positive and negative informational events to occur, which shift a consumer’s score up or down, enabling a more accurate measure of risk and so a greater dispersion in credit scores. Figure 1 shows that immigrants’ higher credit quality extends across the age-40 credit score distribution. For example, the 25th percentile is 621 for immigrants and 566 for non-immigrants. Appendix Figures A8 and A9 show similar patterns across ages and SSN Age cohorts: immigrants are less likely to be subprime (credit score < 600) and, by their mid-to-late thirties, more likely to be prime plus (credit score > 720) or superprime (credit score > 780). Appendix Figure A8 also shows that these findings hold regardless of immigration age or whether we condition on having a score. Panels A and B of Appendix Figure A9 show that they also hold for consumers’ first observed scores, despite the limited information available to construct these.

3.2.4 CREDIT DELINQUENCY & CREDIT UTILIZATION

Immigrants, and later-in-life immigrants, are more creditworthy using two alternative measures that are common inputs to credit scores (Gibbs et al., 2025): any delinquency 90 or more days past due (conditional on holding any credit product), and the average credit card utilization rate (the ratio of the sum of credit card balances across cards to the sum of credit card limits across cards, conditional on holding any credit card). Immigrants have lower delinquency rates (Panel E) and lower credit card utilization (Panel F) than SSN Age 18 consumers. Among immigrants, those arriving later in life generally have lower delinquency rates and utilization.

A complementary way to assess immigrants’ creditworthiness is the “calibration bias” approach used in Bakker et al. (2026). Panel A of Figure 3 shows the delinquency rates for immigrants and non-immigrants by age 37, conditional on credit scores at age 30 for consumers with a credit score by age 30. Across all credit score groups, immigrants have lower average delinquency rates than non-immigrants. Panel A of Appendix Table A4 shows regression estimates, where the outcome is any delinquency by age 37 and regressions include fixed effects for birth year, credit score at age 30, and ZIP5 at age 30. By controlling for credit score at age 30, this accounts for delinquencies by this age. Column 1 of this table shows that immigrants’ delinquency rates are 1.5 percentage points lower than those for non-immigrants, or 5.5% of the non-immigrant mean (27.2%), and columns 3 to 12 show that the differences are most pronounced among consumers with lower credit scores at age 30. Thus, even conditional on credit scores at age 30, immigrants are less likely to become delinquent, suggesting that the limited credit information available at that age may not

fully capture immigrants’ creditworthiness relative to non-immigrants.

3.3 ACCESS TO MORTGAGES AND AUTO LOANS

We now consider the timing of first credit access to mortgages and auto loans, and in the following subsection credit cards. We observe whether consumers access these products, and later discuss whether these equilibrium outcomes are driven by demand or supply factors. The majority of houses and vehicles in the US are purchased on credit and therefore are observed in credit reporting data (Gibbs et al., 2025).

Studying multiple credit products is informative as consumer credit markets vary in their reliance on credit scores and on information in a credit report, including the length of credit history. Credit card decisions are typically more reliant on credit scores than auto loan or mortgage decisions (Blattner et al., 2022). Credit card lending decisions are primarily based on scores (e.g., Agarwal et al., 2018). Whereas for auto loans, the lender may evaluate other information, such as income, especially if a consumer does not have a very high credit score (e.g., Grunewald et al., 2026). For mortgages, Fannie Mae and Freddie Mac eligibility requirements depend not only on credit scores, but also on information obtained directly from the borrower, such as loan-to-value and debt-to-income ratios calculated using verified income information, and the liquid assets available to a consumer after closing.

Despite greater creditworthiness, immigrant credit access is lower than that of non-immigrants. In Figure 4, we show the lifecycle of credit access for each type of credit by SSN Age cohort for each age. Immigrants are less likely than non-immigrants to access credit at any point in the lifecycle. We find that immigrants are *persistently* less likely to ever had access auto loan credit (Panel A) and mortgages (Panel B), even up to age 37, the end of the panel. Moreover, for auto loans and mortgages, there is a larger difference, relative to the SSN Age 18 non-immigrant benchmark, for immigrants arriving later in their twenties than for immigrants arriving earlier in life — the lines do not converge. This marked pattern appears consistent with entry into the credit system later-in-life delaying credit access in ways beyond observable measures of creditworthiness. That is, even though later in life immigrants have significantly higher average credit scores, their access to mortgages and auto loans lags significantly behind non-immigrants and even behind immigrants with an earlier age at immigration.

We use our regression specifications to examine the age of first credit access, separately for each

type of credit, as presented in Panel A of Table 4.⁹ We find that immigrants first access credit significantly later (or not at all) than non-immigrants: translating into 5.0 years later for auto loans (column 1) and 2.3 years later for mortgages (column 3), on average. These average differences grow for later in life immigrants. One additional year of SSN Age predicts an additional delay of 0.6 years for a consumer’s first auto loan (column 2), and 0.4 years for a consumer’s first mortgage (column 4). Appendix Table A2 shows that these estimated delays in first accessing credit are robust to controlling for the consumer’s last ZIP5, longest ZIP5, and number of ZIP5s.

To understand the extensive margin, we change the outcome to whether a consumer has accessed a specific credit product (credit card, auto loan, or mortgage) by age of 37 (i.e., 8 years after the last SSN Age cohort arrives). Panel B of Table 4 presents the results showing large and significant differences in average long-run access to auto loans and mortgages: immigrants are 15.7 percentage points (20.6% of the outcome mean) less likely to have ever had an auto loan (column 1) and 8.9 percentage points (20.0%) less likely to have ever had a mortgage (column 3) by age 37. Columns 2 and 4 allow for heterogeneity in SSN Age and reveal how the lower use of auto loans and mortgages for immigrants, compared to non-immigrants, is substantially larger for immigrants who arrive later in life. Specifically, each additional year of SSN Age is associated with 1.6 pp lower auto loan access and 1.8 pp lower mortgage access by age 37. For mortgages, adding SSN Age reduces the baseline immigrant coefficient from 8.9 to 1.1 pp, so immigration timing accounts for nearly all of the mortgage gap; for auto loans, a baseline gap of 8.7 pp remains. These results are robust to including additional fixed effects (Appendix Table A2), including for credit score at age 30 as shown in Panels B and C of Figure 3 (regression estimates in Panel B of Appendix Table A4). While credit scores at age 30 may partly reflect prior credit access, we show that the gaps persist even among the subsample of consumers who achieve identical observed creditworthiness.

The difference in mortgage access between immigrants and non-immigrants in Panels A and B is largely explained by the timing of immigrants’ arrival in the United States, with lower access emerging for consumers with later in life SSN Ages (see Appendix Figure A6 for heterogeneity by each SSN Age) and controlling for the last observed geographic location (Appendix Table A2). A large immigrant versus non-immigrant difference remains in auto loan access, which we explore in Section 5 by studying ACS and SCF data.

⁹To handle missing values of age at which a consumer first takes out a credit product, we assign these observations a value that corresponds to accessing the product in 2025, one year after the end of our data. We use this approach because excluding such observations from the estimation would understate any difference in credit access.

3.4 ACCESS TO CREDIT CARDS

We now examine consumers' access to credit cards, a key source of unsecured borrowing. Panel C of Figure 4 shows that immigrants are less likely than the SSN Age 18 non-immigrant benchmark to have had any credit card, although this difference largely disappears over the life cycle. We use our regression specifications to study the extensive margin of credit card access, with outcomes being an indicator for the age of first accessing a credit card, and whether a consumer has ever held a credit card by age 37. Estimates are reported in Panel A of Table 4, where we find a significant immigrant to non-immigrant difference in the average age of accessing credit cards: translating into 4.8 years later for immigrants first taking out a credit card (or not at all) in column 5, and the difference is greater for later in life immigrants (column 6). By age 37, these differences in credit card access have largely disappeared: Panel B of Table 4 shows that immigrants are only slightly, 1.4 percentage points, less likely to have a credit card than non-immigrants (column 5) with an economically small but significant difference by SSN Age (column 6).

Panel D of Figure 4 shows the number of credit cards, Panel E shows the total value of credit limits (with additional conditional measures in Panels C and D of Appendix Figure A7), and Panel F shows the credit limit per card conditional on holding any card. Consistent with the evidence on access to auto loans and mortgages, credit card limits of immigrants with later in life SSN Ages take many years to catch up to immigrants with earlier in life SSN Ages and non-immigrants. This slow convergence reflects two different dynamics. The average credit limit per card *never* catches up to earlier cohorts (Panel F). By contrast, the number of credit cards catches up to and surpasses the SSN Age 18 cohort within five years of immigration (Panel D). Additional regression results are in Panel B of Appendix Table A3; Panels D to F of Figure 3 show results controlling for credit score at age 30 (regression estimates in Panel B of Appendix Table A4).

4. PAIRED COHORTS

4.1 METHODOLOGY

Immigrants and non-immigrants differ along many dimensions. To account for this, we use a *paired cohort* strategy, which focuses comparisons on two immigrant cohorts, who are the same age but arrive in the United States a single year apart. This estimation approach is not a causal design, but allows us to quantify how immigrating one year later in life correlates with credit outcomes.

While we have so far defined a cohort as all consumers with the same SSN age, in this section

we define a cohort more narrowly, adding the requirement that all consumers with the same SSN Age are also the same age. Thus, we define a cohort k as the set of consumers who have the same birth year *and* the same year of SSN assignment—e.g., the set of people born in 1985 and who also immigrated in 2007 and were therefore assigned an SSN at age 22 is a cohort in this analysis (Birth Year 1985 $\times$ SSN Age 22). We aggregate the data such that there is one observation per cohort and year from 2000 to 2024. We restrict to focal cohorts with SSN Age ≥ 22 and $2003 \leq \text{Birth Year} + \text{SSN Age} \leq 2011$, ensuring a pre-period and an available control.

A pair p is a set of two cohorts $\{k', k''\}$, consisting of a focal cohort k' and a matched control cohort k'' with the same birth year but SSN Age one year earlier in life: $\text{SSN Age}_{k''} = \text{SSN Age}_{k'} - 1$. There are 68 pairs of cohorts containing consumers who had their SSNs assigned between 2003 and 2011, and we trim the data to ensure that our panel is balanced within and across pairs. We define event time t based on the focal cohort, keeping 16 years of annual data from two years prior to the year when the focal cohort's Age = SSN Age, to thirteen years after. Our dataset has a paired structure with 2,176 cohort-by-year observations, weighted by cohort size.

We estimate the following specification via OLS:

$$Y_{k(p),t} = \sum_{\tau=-1}^{+13} \delta_{\tau} (\mathbb{I}\{\text{focal cohort}\}_{k(p)} \times \mathbb{I}\{t = \tau\}) + \pi_{k(p)} + \nu_{p,t} + \epsilon_{k(p),t} \quad (3)$$

where $Y_{k(p),t}$ is the outcome variable in event time t for cohort k belonging to pair p . This is a stacked difference-in-differences design, with $\mathbb{I}\{\text{focal cohort}\}_{k(p)}$ as the treatment indicator and $\tau = -2$ as the omitted period. δ_{τ} therefore shows how credit outcomes, observed in the same year within a pair, develop for each cohort relative to consumers with the same birth year who immigrated one year earlier. Cohort fixed effects $\pi_{k(p)}$ and pair-by-event-time fixed effects $\nu_{p,t}$ absorb level differences and pair-specific trends, isolating variation within a pair. We calculate 95% confidence intervals using wild cluster bootstrapping (Webb weights, 10,000 iterations), clustered by birth year. This clustering allows for dependence over the lifecycle and within pairs—important in our context because a cohort can serve as the focal cohort in one pair and as a control in another.

4.2 RESULTS

We observe distinct dynamics for credit market entry versus later-in-lifecycle credit access using the paired cohort approach. Figure 5 presents the results on accessing any credit product or credit card (Panel A), access to auto loans and mortgages (Panel B), and credit card limits (Panel C).

To emphasize the non-standard timing, we label event time $t = -1$ as “C” and $t = 0$ as “T,” denoting the years of SSN assignment for the control and focal cohorts, respectively. Reinforcing earlier results, we observe a temporary difference in the relationship between one year later in life immigration and accessing credit cards. In the year the focal cohort receives its SSN — event time T — focal immigrants are 20 percentage points less likely to have any credit card relative to control immigrants (same-age immigrants with an SSN Age one year lower). This difference disappears by year four, and results are near-identical for accessing any credit product.

In contrast, access to mortgage and auto loans does not converge. Thirteen years after immigrating, immigrants, relative to same-age immigrants who were assigned an SSN one year earlier in life, are 1.12 percentage points *less* likely to have had an auto loan (95% C.I. -1.33 to -0.92 , control cohort mean at $T + 13$ is 58.1%) and 1.36 percentage points less likely to have had a mortgage (95% C.I. -1.80 to -0.89 , control cohort mean at $T + 13$ is 35.2%). Thus, even a one-year difference in SSN Age correlates with long-run differences in credit access.

Consistent with the idea that later in life immigration cohorts’ initially delayed credit access results in persistently shorter credit histories, we estimate that one year later in life SSN Age is associated with cumulatively 0.73 to 0.78 years less credit history relative to same-age immigrants, and most of this delay is realized by the focal cohort’s event time $T + 1$ (Appendix Table A5 and Figure A10). Ten years after the focal cohort’s immigration, we estimate an average delay of 0.28 years for first mortgages, and a delay of 0.44 years for first auto loans from one year later in life immigration relative to same-age immigrants. Most of the delay cannot be attributed to “mechanical delay” because it emerges substantially after the year of SSN assignment, aligning with when in the lifecycle these products are typically accessed for the first time by non-immigrants.

Panel C of Figure 5 shows that the focal cohort has significantly lower credit card limits ($-\$474$, 95% C.I. $-\$746$, $-\$207$, control cohort mean at $T + 8$ is $\$17,121$) for the first eight years after immigrating. This difference is largest after two years, with average credit limits being $\$1,705$ *lower* than those of the control cohort (95% C.I. $-\$1,936$ to $-\$1,478$, control cohort mean at $T + 2$ is $\$5,418$) and shrinks over time becoming indistinguishable from zero after eight years, with similar results for the number of credit cards (Appendix Figure A10).

5. MECHANISMS

5.1 CREDITWORTHINESS

In Section 3 we found that immigrants—and especially later-in-life immigrants—have higher credit scores and less risky credit behaviors. Higher credit scores would not be expected to result in lower long-term access to auto loans and mortgages, nor delayed access to higher credit card limits. Our calibration bias approach revealed that, conditional on credit score at age 30, immigrants are less likely to become delinquent or access autos or mortgages by age 37. So, measured creditworthiness does not appear to explain our results. The next subsections consider alternative mechanisms: emigration, credit demand, supply, tastes, history dependence, income, and culture.

5.2 EMIGRATION

A different possible explanation for some of our results, especially lower long-term access to mortgages, could be emigration: consumers expecting to emigrate from America may be less likely to purchase property. Even if immigrants and non-immigrants were similarly likely to purchase property each year, emigration would mean that immigrants are present in our data for fewer years, making catch-up less likely. We proxy for emigration based on whether a consumer disappears from our credit data, which may also occur for other reasons, such as death. This approach appears to be a reasonable proxy for emigration as immigrants are significantly more likely to exit our data than non-immigrants, as shown in Panel A of Appendix Table A6.

Our main findings remain robust after accounting for this differential attrition of immigrants. In Appendix Figure A11 and Panel B of Appendix Table A6, we condition on consumers still present in the data in 2024 and examine all of the main credit outcomes; the results are consistent with those in our main Figure 4. These are also robust to the emigration proxy used (Appendix Figure A11 and Table A6). Therefore, while emigration may be a factor that influences some credit decisions, it does not fully account for all the differences in credit access that we document.

A related concern is that the possibility of future emigration leads consumers to avoid investing in assets that require financing. While shorter exposure time complicates comparisons, we observe lower—not higher—attrition among immigrants arriving later-in-life. Furthermore, because access gaps persist even among consumers remaining through 2024, emigration does not fully explain our results. See Appendix Table A6 for details.

5.3 CREDIT DEMAND, SUPPLY, AND TASTES

Another possible mechanism driving lower immigrant access to credit is that they have lower credit demand than non-immigrants. We examine this idea directly using data on individual credit inquiries available for 78,365 consumers, a random subset of those contained in the credit data. As a measure of credit demand, we use inquiries following Gibbs et al. (2025). To proxy for credit supply, we infer credit rejections from inquiries that do not result in a new origination within the next 270 days, following Blattner and Nelson (2024). These are imperfect proxies as credit inquiries are one of the final stages in searching for credit, and so may under-state latent demand. Also, a credit inquiry without a resulting origination does not always imply a credit rejection.

Table 5 presents regressions using a panel of consumer-years to relate credit inquiries and rejections to immigration status and timing, with fixed effects for birth year, first three-digit ZIP code (rather than five-digit due to the smaller size of this dataset), and controls for pre-inquiry credit score. Columns (1) and (2) show that immigrants are 1.7 percentage points less likely to apply for an auto loan in a given year (relative to the non-immigrant mean of 12.7%), consistent with lower demand for auto financing. The mortgage inquiry gap is much smaller and not significant once we allow for heterogeneity by immigration age (column 4), suggesting that lower inquiry rates are specific to auto loans rather than a general pattern. Crucially, the SSN Age coefficient is not significant for either auto loan or mortgage inquiries. Immigrants who arrived later in life do not apply for credit at different rates than those who arrived earlier in life, so these demand differences do not appear to explain differences in credit outcomes by age of immigration.

Columns (5) and (6) show rejection rates (conditional on any inquiry). When immigrants apply for credit, they face significantly higher rejection rates: 3.7 percentage points for auto loans (column 5, non-immigrant mean is 38.8%) and 2.1 percentage points for mortgages (column 6, non-immigrant mean is 55.2%).¹⁰ These rejection rates are consistent with information frictions or supply-side constraints. We are underpowered to estimate the SSN Age specification because only 6% (mortgages) and 12% (autos) of the consumer-year observations contain inquiries.

Differences in asset-linked credit – auto loans and mortgages – could also reflect different tastes for the underlying asset. That is, immigrants could value cars less than non-immigrants or be more likely to prefer renting over owning. To examine this, we use repeated cross-sections from the American Community Survey (ACS), which allow us to observe asset ownership directly — cars

¹⁰Panels E to H of Appendix Figure A7 and Panel D of Figure A10 use less granular inquiries data to show the difference reverse in immigrants' thirties, potentially consistent with being deterred by earlier rejections.

and homes — and to control flexibly for detailed demographics, including income.

Figure 6 plots estimates from regressions (with demographic controls) of mortgage status (Panel A), homeownership (Panel B) and car ownership (Panel C) on immigrant status across four waves of the ACS spanning 2005 to 2023. Immigrants arriving in their twenties are significantly and persistently less likely to have a mortgage. Adding controls for log household income, employment, and occupation attenuates the estimates only slightly. Immigrants’ mortgage gap narrows as consumers accumulate more time in America. Patterns in homeownership are similar, consistent with Borjas (2002). Our results are consistent with mortgage credit often being necessary to become a homeowner and rule out that the mortgage gaps reflect immigrants being wealthier and purchasing property outright with cash. The ACS does not record auto loan use. We observe gaps in auto ownership that appear initially but do not persist. Because the ACS captures household-level asset ownership, differences are not explained by non-immigrant partners of immigrants owning assets.

Differences in asset-linked credit potentially reflect immigrants’ different tastes for financing cars or homes. We investigate this using the 2022 Survey of Consumer Finances (SCF), the only vintage that identifies immigrants. Appendix Table A9 shows that immigrants are 15 percentage points less likely than non-immigrants (80% of whom agree) to view borrowing to buy a car as acceptable, and this attitude strongly predicts auto loan use. The SCF has no analogous question about attitudes to credit card or mortgage debt. A question on general attitudes toward debt (columns 7 and 8) shows no clear evidence of a more general immigrant “debt aversion” (e.g., Martínez-Marquina and Shi, 2024), and other SCF results lack statistical power (Appendix Table A10).

5.4 HISTORY DEPENDENCE, INCOME, AND CULTURE

A plausible mechanism for our findings is that the length of a consumer’s credit history itself may be an important factor in lending decisions, as immigrants arrive with no credit history and cannot be scored, while later-in-life immigration necessarily reduces the length of credit history. This mechanism yields two predictions: among same-age immigrants, credit access should decline with age at arrival, and the decline should be steeper for auto loans and mortgages than for holding a credit card, since card issuers rely primarily on credit scores. Both hold. The paired cohort analysis shows that the auto loan and mortgage differences persist for more than a decade.

This mechanism is consistent with our evidence of lower immigrant credit access despite higher creditworthiness than same-age non-immigrants. History dependence of credit is consistent with the differences in credit access observed in the paired cohort analysis.

To the extent that having no or a limited credit history is a financial friction for immigrants that rations credit, market solutions are emerging to expand credit access. Research suggests that credit access can be expanded to consumers without credit scores by lenders using alternative data (e.g., Chioda et al., 2026, Di Maggio and Ratnadiwakara, 2026). American Express has linked their internal data internationally, allowing their foreign customers to open a U.S. American Express card. FinTech lenders specialized in lending to immigrants have recently launched, such as LendBuzz for auto loans, and America Mortgages. FinTech lenders, such as MPower Financing and Prodigy Finance, provide finance for the cost of educating foreign students at select American universities (Bloomberg, 2025). FinTech firms (e.g., Nova Credit) and credit reporting agencies have recently developed technological solutions to enable access to foreign credit reports and to translate foreign credit scores for lenders (FinTech Magazine, 2024, TransUnion, 2018, Equifax, 2024).

We cannot be certain that the key mechanism for delayed immigrant credit access is only the number of years in the United States credit system, because there are also other sources of history dependence. Our results are organized around life cycles defined by consumers’ birth years, however, it is possible that immigrants’ life cycles are instead more closely linked to when they arrive in the United States — when their American labor market and immigrant experience begins — which is a different type of history dependence. Being in the country for an additional year means that the paired control cohort has an additional year to potentially take out credit, an extra year of U.S. earnings, and is a year further along their personal and professional American life cycle, which, in turn, may mean they are a year further along in the credit demand life cycle.

We examine history dependence in earning ability in the ACS. After controlling for demographics, immigrants arriving in their twenties have substantially lower household income (11% in 2005-2013 and 4% in 2019-2023) than non-immigrants (Appendix Table A7, this also holds unconditionally: Appendix Figure A1). This gap is larger for those who immigrate later in their lifecycle (broadly consistent with Chiswick, 1978 and Kerr et al., 2026). These income differences may contribute to lower access to asset-linked credit, either through demand or supply. Figure 6 and Appendix Table A8 show that even after controlling for income and other factors, persistent differences in outcomes remain. Each additional year of later-in-life immigration leads to lower mortgage use, homeownership, and, to a lesser extent, car ownership. This points toward additional frictions beyond earnings capacity. Differences being largest earlier in the life cycle is also potentially consistent with financial frictions, as this is when consumers are most credit-dependent.

One possibility other than history dependence is that the lower credit outcomes by later-in-life

immigration could instead reflect cultural differences correlated with country of origin and timing of immigration. We study this source of heterogeneity by estimating regressions that include non-immigrants and restrict to subsamples defined by region of origin, country of origin, and terciles of home country homeownership rates calculated from data in Gorback and Schubert (2026).

Irrespective of country of origin, immigrants typically have lower mortgage use than non-immigrants. Estimates vary in size, likely reflecting cultural heterogeneity (Appendix Figure A12). Studying within origin variation in immigration age helps separate arrival timing from broad cultural differences. In Panel A of Appendix Figure A13, the immigration age coefficient is negative and statistically significant in every region of origin group in every wave. Estimates attenuate as the immigration cohorts accumulate more time in the U.S. Panel B shows a consistent pattern within each country of origin, as does Panel C within all terciles of home country homeownership rates. Therefore, history dependence remains pervasive across cultures.

6. CONCLUSION

This study offers large-scale evidence on immigrants’ assimilation into the U.S. consumer credit system. We document how immigrants’ creditworthiness and credit usage evolve with age, comparing immigrants both to non-immigrants and to immigrants who arrived later in life. Upon credit market entry, immigrants are positively selected on credit quality: credit scores are approximately 25 points higher than non-immigrants’ throughout their thirties. Yet by age 37, immigrants are 16 and 9 percentage points less likely to have ever had an auto loan or a mortgage than non-immigrants of the same age and geographic area.

Among same-age immigrants, arriving just one year later is associated with persistently lower auto loan and mortgage use. We argue that this reflects history dependence: credit access depends on time in the U.S., beyond what credit scores capture. An immigrant’s age at arrival matters little for credit cards, where underwriting is primarily based on credit scores, but matters for a decade or more for auto loans and mortgages, where lenders draw on broader information that accumulates with time in the country. These patterns survive controls for income, hold within country of origin, and are not explained by differences in demand or emigration. In light of the major contributions of immigrants in credit-dependent areas such as entrepreneurship (Bernstein et al., 2025), our results suggest an opportunity for innovations to expand lending to immigrants in the United States.

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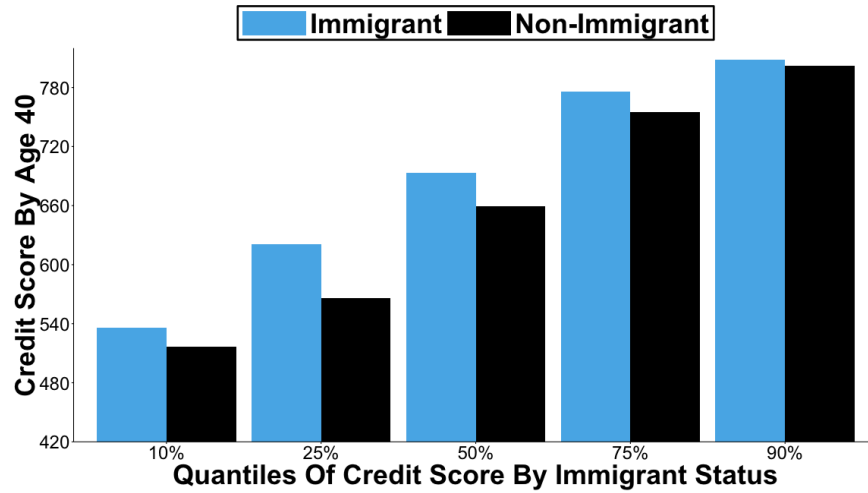
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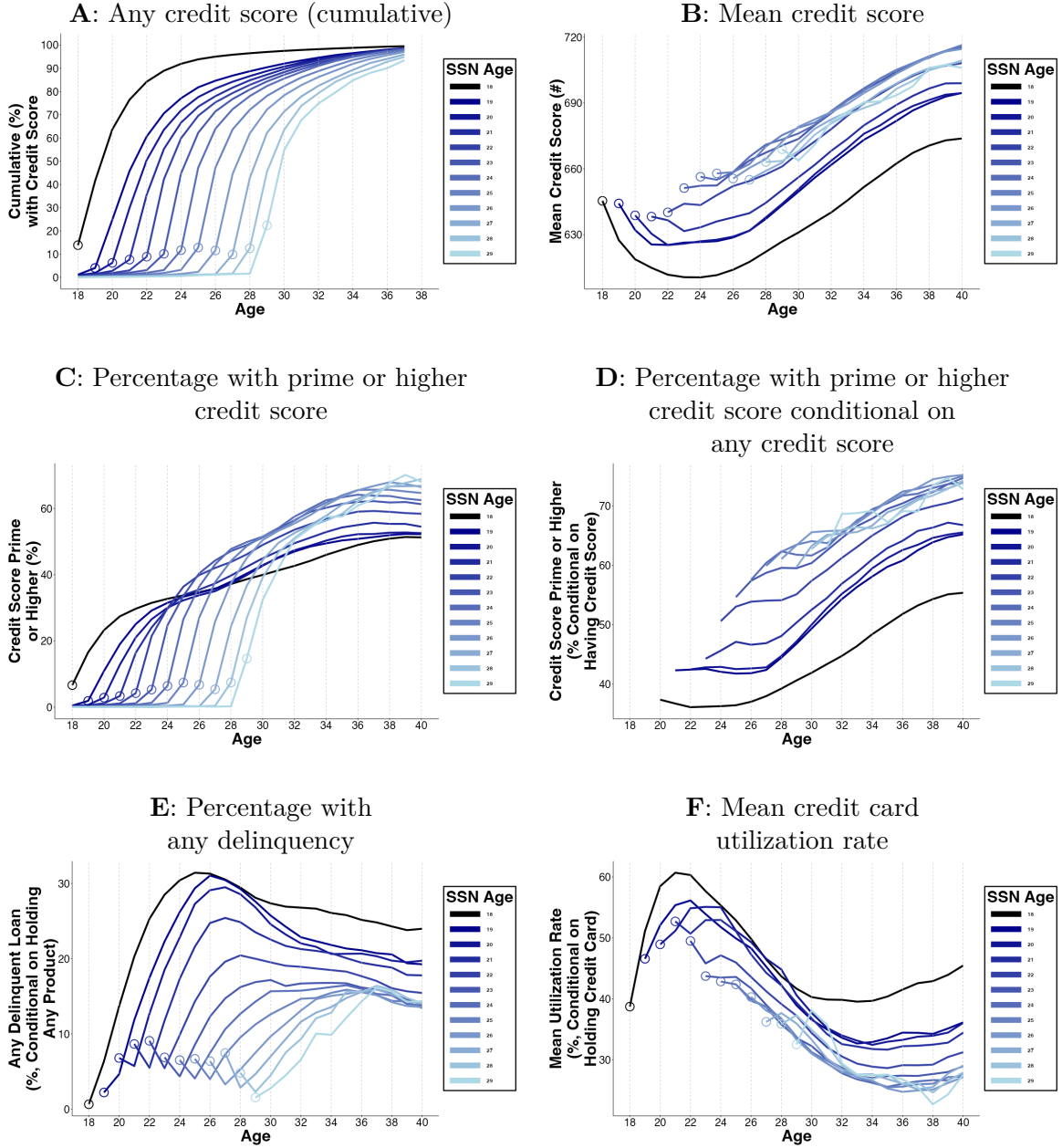
FIGURES AND TABLES

Figure 1: Quantiles of credit scores by age 40 by immigration status



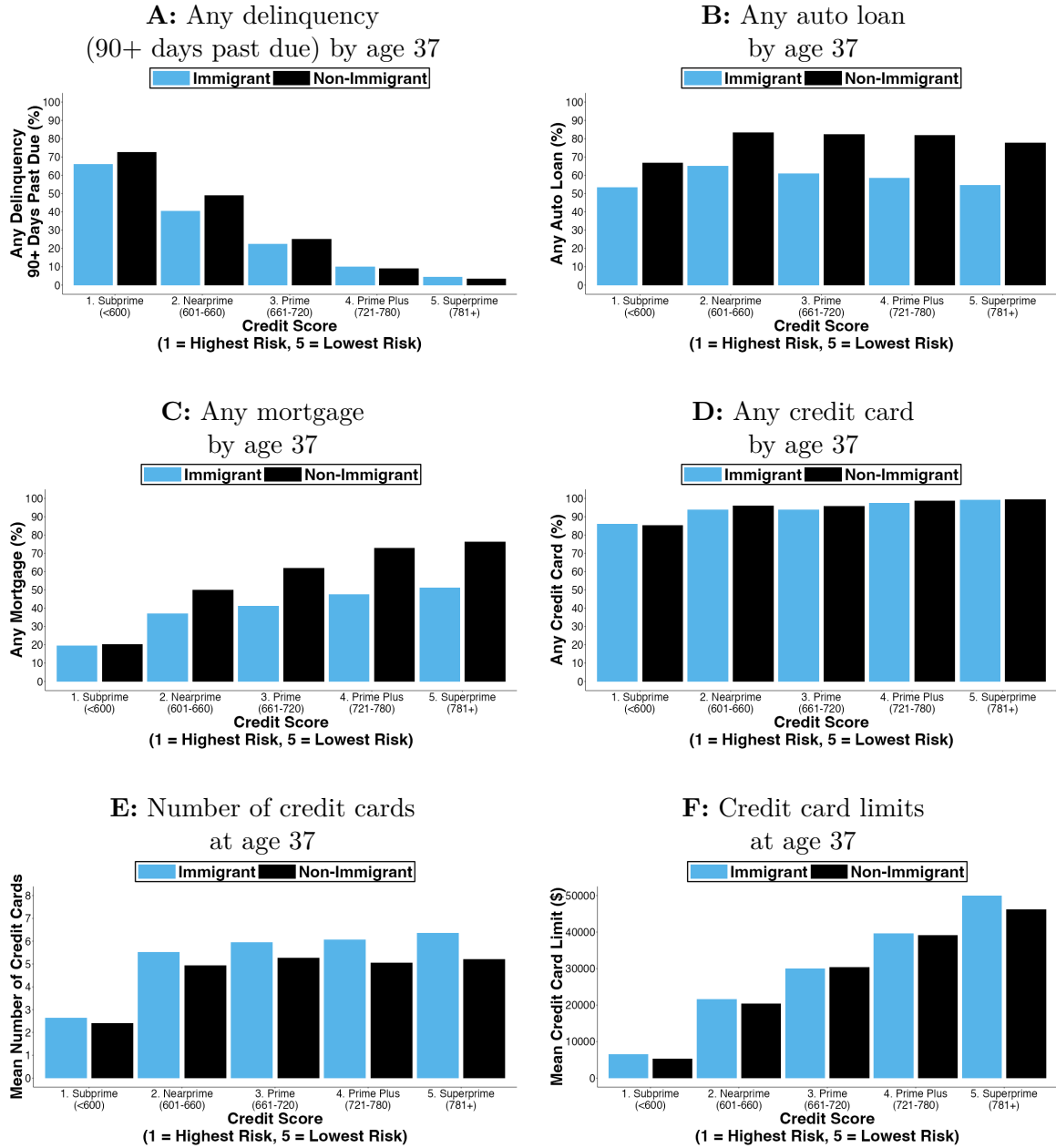
This figure shows the values of credit scores by age 40 taken at the 10%, 25%, 50%, 75%, and 90% quantiles of the distribution. These quantiles are calculated separately for immigrants and non-immigrants. We classify immigrants as those assigned SSNs at ages 21 to 29, and non-immigrants as those assigned SSNs at age 20 or younger. The data consist of 6,252,685 consumers with non-missing credit scores by age 40, of which 376,461 are immigrants and 5,876,224 are non-immigrants. For consumers who are not observed in the data at age 40, this uses their credit score at the oldest age before 40 that is observed.

Figure 2: Lifecycle of creditworthiness by SSN Age cohorts



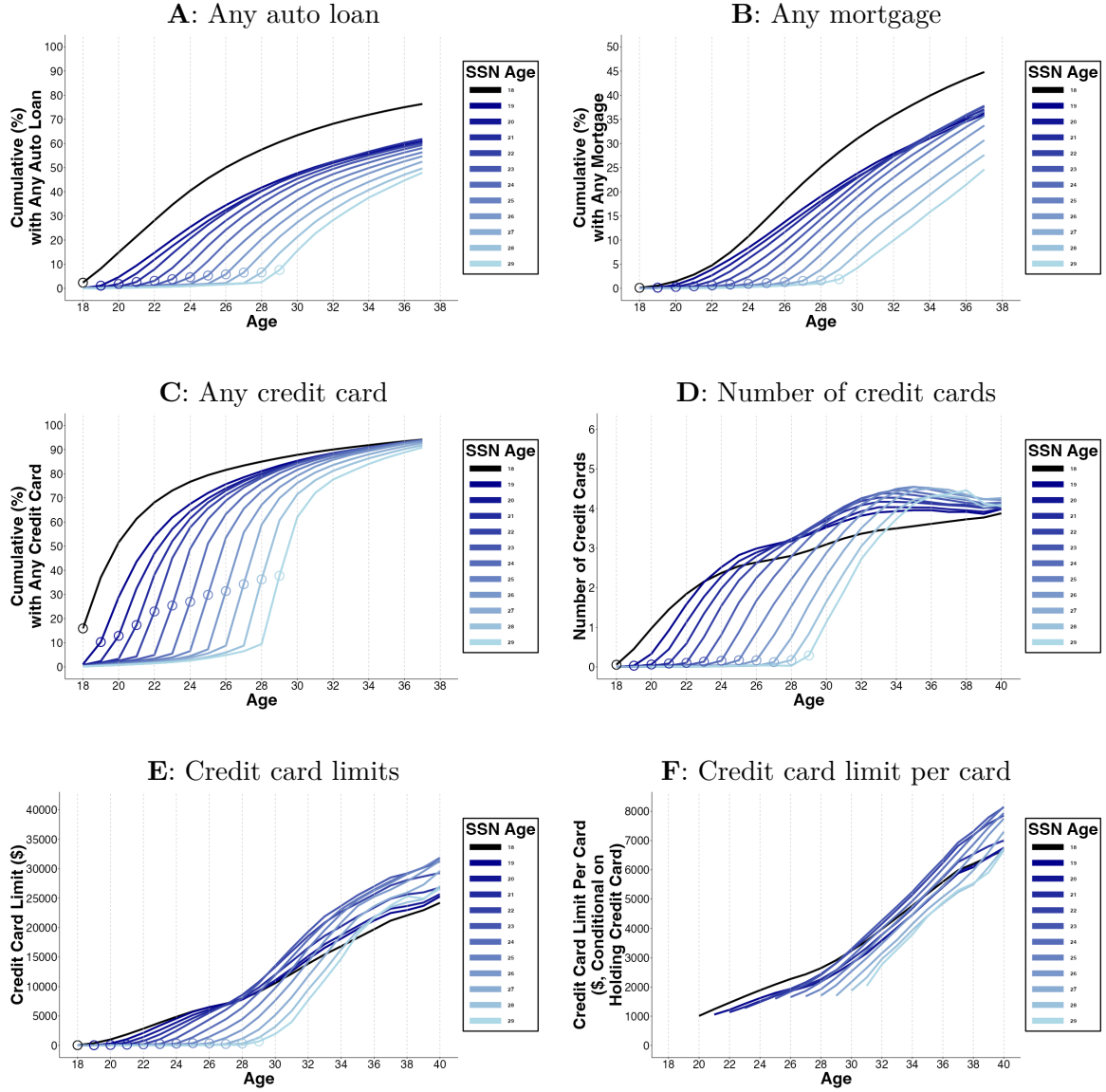
This figure plots outcomes at different calendar ages for each SSN Age cohort. The black line (SSN Age 18) pools all consumers with SSN Age 18 or younger. Lighter colors indicate later in life SSN Age cohorts from 19 to 29. We classify immigrants as the SSN Age cohorts 21 to 29. $Age = SSNAge$ is indicated by the circles on each line. This figure uses data for birth years from 1982 to 1987. Because Panel A is a cumulative chart, we end it at age 37, whereas for the other panels the estimates for ages 38 to 40 account for this attrition. Consumers with the birth years 1985, 1986, and 1987, are not observed for age 40, ages 39 to 40, and ages 38 to 40, respectively, by the end of our data in 2024. Panel A presents the cumulative fraction of consumers that have been credit scored by each age. In Panels C and D, Prime or Higher Credit Score is a credit score of 661 or higher. Panel E is conditional on holding any credit product; Panel F is conditional on holding a credit card.

Figure 3: Credit outcomes by age 37 conditional on credit score at age 30



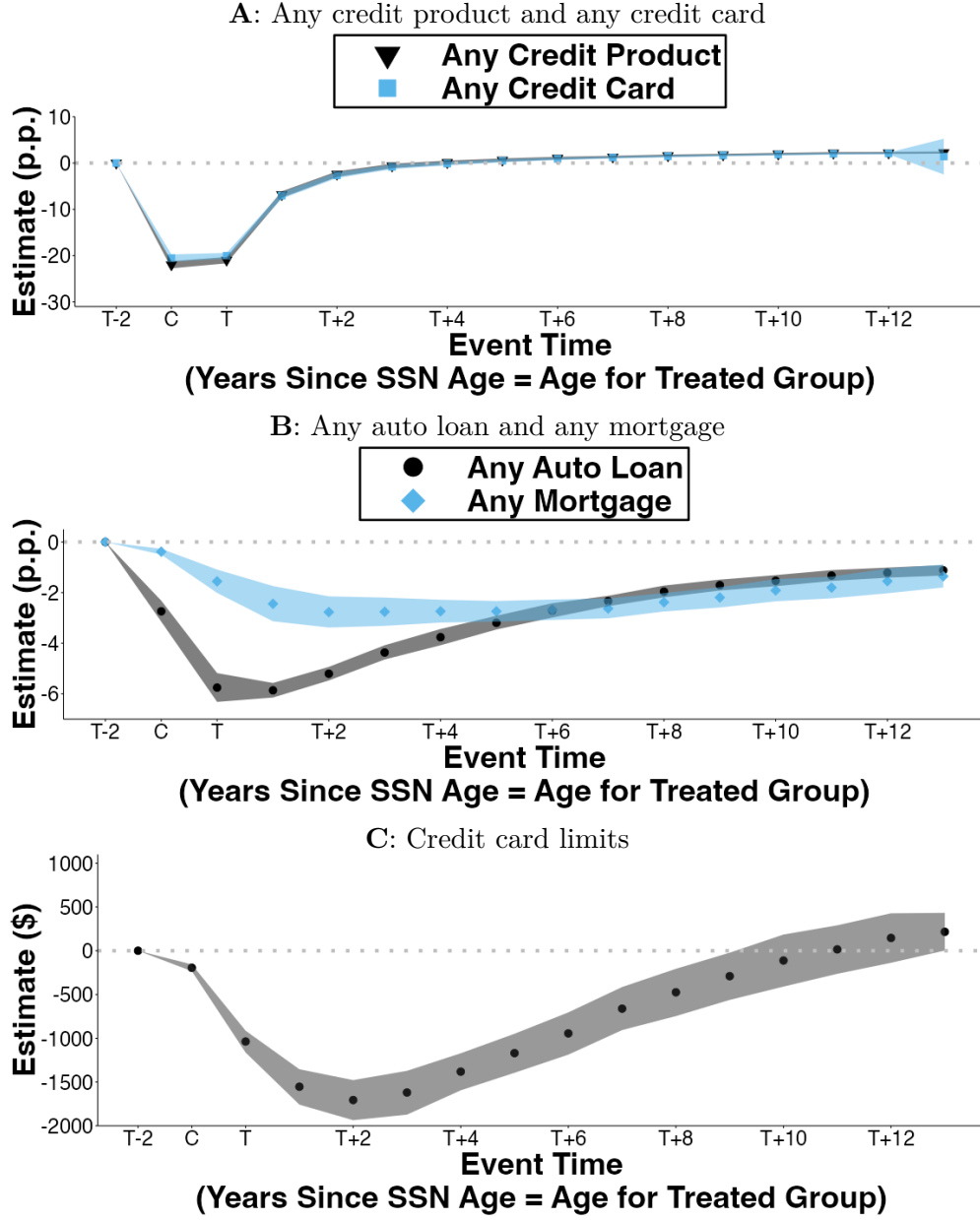
Each panel shows mean outcomes measured at age 37 conditional on a consumer's credit score measured at age 30. Each panel splits results by immigrants and non-immigrants, where immigrants are those with a SSN Age of 21+.

Figure 4: Lifecycle of credit access by SSN Age cohorts



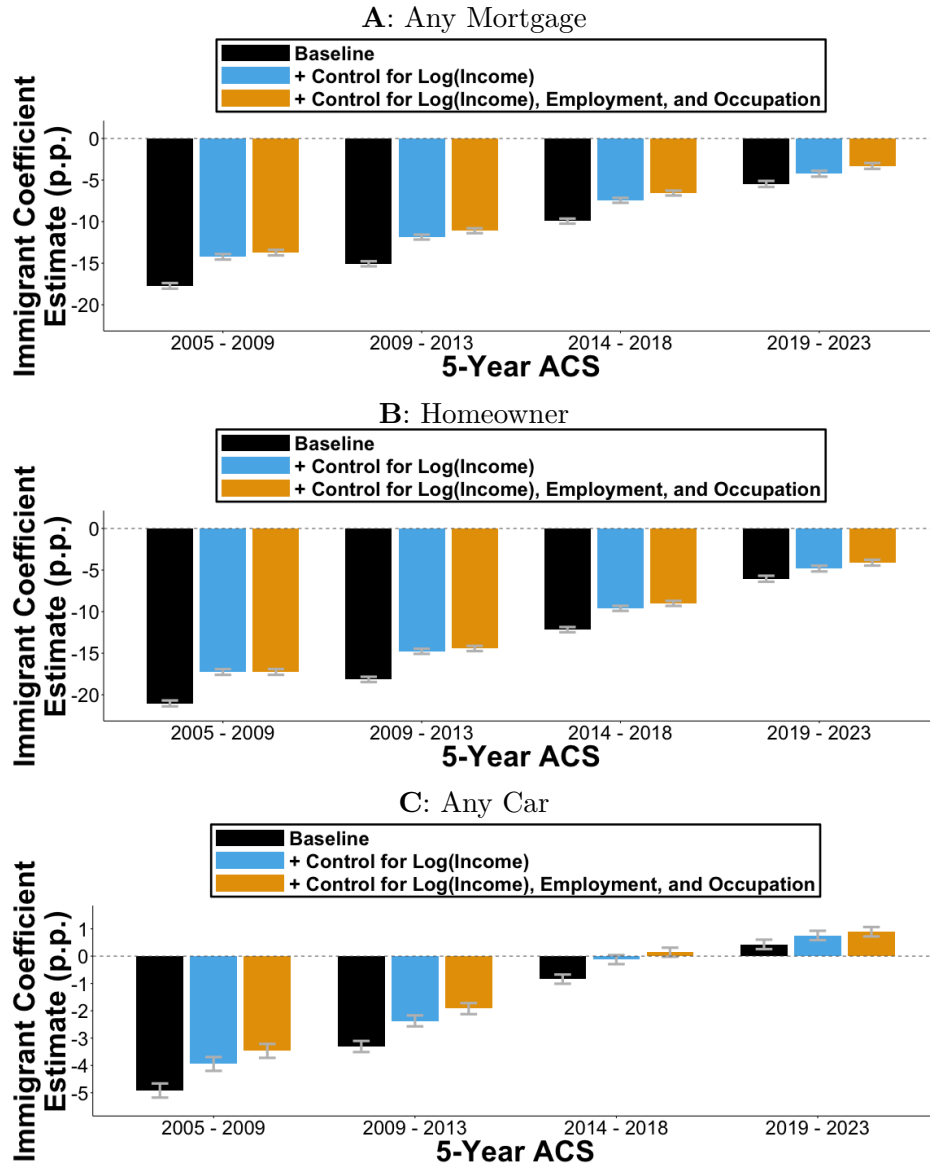
This figure plots credit outcomes at different calendar ages for each SSN Age cohort. The black line (SSN Age 18) pools all consumers with SSN Age 18 or younger. Lighter colors indicate later in life SSN Age cohorts from 19 to 29. We classify immigrants as the SSN Age cohorts 21 to 29. $Age = SSNAge$ is indicated by the circles on each line. Panels A, B, and C use data for birth years from 1975 to 1987. Because Panels A, B, and C are cumulative charts, we end them at age 37, whereas for the other panels the estimates for ages 38 to 40 account for this attrition. Consumers with the birth years 1985, 1986, and 1987, are not observed for age 40, ages 39 to 40, and ages 38 to 40, respectively, by the end of our data in 2024. Panels D, E, and F use data for birth years from 1982 to 1987 and plot means of the outcomes.

Figure 5: Paired cohorts: Dynamics of credit access



This figure presents dynamic estimates for differences in credit access between a cohort with SSN Age = s and a matched same-age control cohort with SSN Age = $s - 1$. The differences are presented in event time, where C is the year when age equals SSN Age for the $s - 1$ (control) cohort and event time T is the year when age equals SSN Age for the s (treatment) cohort. Panels A and B show estimates, in percentage points, from regressions where the binary outcomes measure any credit product (black) and any credit card (blue) in Panel A, and in Panel B, any auto loan (black) and any mortgage (blue). Panel C shows estimates for the outcome, the value credit card limits (\$). The shaded areas indicate 95% confidence intervals, calculated using wild cluster bootstrapping by birth year using Webb weights and 10,000 iterations.

Figure 6: American Community Survey: Immigrant coefficient estimates on mortgage use, homeowner, and car ownership



This figure uses public data from the 5-year American Community Survey (ACS) waves 2005-2009, 2009-2013, 2014-2018, and 2019-2023, keeping consumers with birth years 1975 to 1987, non-zero income, and immigrants who immigrated between ages 21 and 29 along with non-immigrants. Each bar shows estimates from a separate cross-sectional regression. Each bar shows coefficient estimates on an indicator for immigration status (*Immigrant* is an indicator for whether the consumer's age of immigration was at age 21 or older). Each x-axis category shows results from a different 5-year ACS wave. The outcome in Panel A is whether the consumer has a mortgage, in Panel B it is whether they are a homeowner, and in Panel C is whether they have an auto. All regressions include fixed effects for the consumer's birth year, geographic area (Consistent Public Use Microdata Areas), sex, education levels, race, hispanic, and survey year. The blue bars also include a control for Log (Income), the log of household income that is winsorized at 99.9th percentile and is scaled to mean zero and standard deviation one. The orange bars additionally include fixed effects for employment status and for occupation. Units are percentage points. 95% confidence intervals are calculated using heteroskedasticity-robust standard errors.

Table 1: Summary statistics on immigrants and non-immigrants

This table presents means and counts for the number of consumers (N) these are calculated from. Means are calculated separately for non-immigrants ($SSN\ Age < 21$) and immigrants ($SSN\ Age\ 21\ to\ 29$) in our data. $SSN\ Age$ is the age that a consumer is assigned an SSN. The variables *Credit Score by Age 30* and *Credit Score by Age 40* are conditional on having a non-missing credit score by ages 30 and 40 respectively (for consumers not observed at a given age, we use the score at the oldest age observed before that age). The variables *Any Score by Age 30* and *Any Score by Age 40* are whether a consumer had any score by those ages (for consumers not observed at a given age, we use whether they had a score at the oldest age observed before that age). The variables *Any Auto Loan*, *Any Mortgage*, and *Any Credit Card* indicate whether, up to the end of our data in 2024, the consumer is ever observed to have any auto loan, mortgage, or credit card respectively. The variables *Age at First Auto Loan*, *Age at First Mortgage*, and *Age at First Credit Card* are each conditional on the consumer ever having auto loans, mortgages, credit cards respectively at any point up to the end of our data in 2024. The variable *Credit Card Limits at Age 40* has smaller sample sizes because birth years 1985, 1986, and 1987 are not observed at age 40.

	Non-Immigrants (<i>SSN Age < 21</i>)		Immigrants (<i>SSN Age 21 to 29</i>)	
	Mean	N	Mean	N
Credit Report				
Age at First Credit Report	20.19	5,914,891	26.09	384,396
Credit Score				
Any Score by Age 30 (%)	96.26	5,914,891	81.63	384,396
Credit Score by Age 30	624.2	5,693,584	658.3	313,799
Any Score by Age 40 (%)	99.35	5,914,891	97.94	384,396
Credit Score by Age 40	658.1	5,876,224	687.5	376,461
Auto Loan				
Any Auto Loan (%)	81.14	5,914,891	65.62	384,396
Age at First Auto Loan	26.13	4,799,057	30.59	252,243
Mortgage				
Any Mortgage (%)	50.28	5,914,891	44.42	384,396
Age at First Mortgage	29.53	2,973,846	32.88	170,754
Credit Card				
Any Credit Card (%)	96.87	5,914,891	97.39	384,396
Age at First Credit Card	22.33	5,730,022	27.20	374,365
Credit Card Limits at Age 30	\$11,492	5,914,891	\$10,211	384,396
Credit Card Limits at Age 40	\$22,112	4,579,356	\$26,608	342,258

Table 2: Timing of credit market entry

Columns 1 and 3 in this table are estimates from the cross-sectional OLS regression specified in Equation 1 that includes an indicator for immigration status (*Immigrant* equals 1 if the consumer's SSN was assigned at age 21 or older). Columns 2 and 4 in this table are estimates from the OLS regression specified in Equation 2 that contains both the indicator for immigration status, and also *SSN Age*, the consumer's age at SSN assignment (pooling consumers with SSN Age 20 or younger into one group assigned a value of 20). All regressions include fixed effects for: the birth year of the consumer and their first observed ZIP code (First Zip5). The outcome in columns 1 and 2 is age at first credit report, and the outcome in columns 3 and 4 is age at first credit product. In these regressions, consumers who never have a credit report or never have a credit product are assigned their age as of 2025. The units in this table are years of age. Standard errors are clustered by birth year $\times$ the first two digits of a consumer's first observed ZIP code. * $p < .05$; ** $p < .01$; *** $p < .005$.

Dep Var: Age at First...	Credit Report		Credit Product	
	(1)	(2)	(3)	(4)
Immigrant	5.53*** (0.02)	2.29*** (0.02)	5.20*** (0.02)	2.07*** (0.03)
SSN Age		0.75*** (0.01)		0.73*** (0.01)
F.E. Birth Year	X	X	X	X
F.E. First Zip5	X	X	X	X
R^2	0.193	0.209	0.118	0.126
N	6,297,340	6,297,340	6,297,340	6,297,340
Mean, SSN Age <21	20.19	20.19	21.56	21.56

Table 3: Creditworthiness

As in Table 2, this table reports estimates of equations 1 and 2. *Immigrant* is an indicator for whether the consumer's SSN was assigned at age 21 or older; *SSN Age* is the consumer's age at SSN assignment (pooling consumers with SSN Age 20 or younger into one group assigned a value of 20). The outcomes in Panel A are average credit score by ages 30 (columns 1 and 2) and 40 (columns 3 and 4). These Panel A outcomes are conditional on having a non-missing credit score by ages 30 and 40 respectively (for consumers not observed at a given age, we use the score at the oldest age observed before that age). The outcomes in Panel B are the probability of prime or higher credit score, measured by a score of 661 or higher, by age 30 (columns 1 and 2) and age 40 (columns 3 and 4), where consumers without a credit score are given a value of zero (for consumers not observed at a given age, we use the score at the oldest age observed before that age to measure whether their score is prime or higher). The units in Panel A are credit score points, and the units in Panel B are percentage points. Standard errors are clustered by birth year $\times$ the first two digits of a consumer's first observed ZIP code. * $p < .05$; ** $p < .01$; *** $p < .005$.

Panel A: Average credit scores

Dep Var: Credit Score by ...	Age 30		Age 40	
	(1)	(2)	(3)	(4)
Immigrant	26.7*** (0.4)	17.2*** (0.6)	24.6*** (0.4)	14.2*** (0.5)
SSN Age		2.3*** (0.1)		2.4*** (0.1)
F.E. Birth Year	X	X	X	X
F.E. First Zip5	X	X	X	X
R^2	0.135	0.135	0.124	0.124
N	6,005,430	6,005,430	6,250,733	6,250,733
Mean, SSN Age <21	624.2	624.2	658.1	658.1

Panel B: Probability of prime or higher credit score

Dep Var: Prime or Higher by ...	Age 30		Age 40	
	(1)	(2)	(3)	(4)
Immigrant	4.61*** (0.17)	6.82*** (0.25)	11.24*** (0.17)	7.04*** (0.22)
SSN Age		-0.51*** (0.05)		0.98*** (0.04)
F.E. Birth Year	X	X	X	X
F.E. First Zip5	X	X	X	X
R^2	0.102	0.102	0.098	0.098
N	6,297,340	6,297,340	6,297,340	6,297,340
Mean, SSN Age <21	37.63	37.63	49.33	49.33

Table 4: Credit access

As in Table 2, this table reports estimates of equations 1 and 2. *Immigrant* is an indicator for whether the consumer's SSN was assigned at age 21 or older; *SSN Age* is the consumer's age at SSN assignment (pooling consumers with SSN Age 20 or younger into one group assigned a value of 20). The outcomes in Panel A are age at first auto loan, age at first mortgage, and age at first credit card (for consumers who do not access a product by the end of our data, we assign them a value that corresponds to accessing that product in 2025, one year after the end of our data). The outcomes in Panel B are the probability of having ever held an auto loan by age 37, held a mortgage by age 37, or held a credit card by age 37. The units in Panel A are years of age and the units in Panel B are percentage points. Standard errors are clustered by birth year $\times$ the first two digits of a consumer's first observed ZIP code. $*p < .05$; $**p < .01$; $***p < .005$.

Panel A: Age of credit products first taken out

Dep Var: Age at First...	Auto Loan		Mortgage		Credit Card	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	4.96*** (0.05)	2.27*** (0.05)	2.26*** (0.04)	0.40*** (0.04)	4.80*** (0.02)	1.79*** (0.03)
SSN Age		0.63*** (0.01)		0.43*** (0.01)		0.70*** (0.01)
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X
R^2	0.083	0.084	0.082	0.083	0.082	0.086
N	6,297,340	6,297,340	6,297,340	6,297,340	6,297,340	6,297,340
Mean, SSN Age <21	29.40	29.40	36.49	36.49	22.99	22.99

Panel B: Probability of having held a credit product by age 37

Dep. Var: By Age 37, has ...	Auto Loan		Mortgage		Credit Card	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	-15.66*** (0.18)	-8.69*** (0.23)	-8.91*** (0.17)	-1.05*** (0.23)	-1.35*** (0.08)	-0.37*** (0.11)
SSN Age		-1.62*** (0.04)		-1.83*** (0.04)		-0.23*** (0.02)
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X
R^2	0.041	0.042	0.063	0.063	0.024	0.024
N	6,297,340	6,297,340	6,297,340	6,297,340	6,297,340	6,297,340
Mean, SSN Age <21	76.06	76.06	44.63	44.63	94.12	94.12

Table 5: Credit inquiries and credit rejections

Each column shows the results of a different regression. Regressions use data for the subset of consumers in the inquiries dataset, which covers 2009 to 2019. Columns (1), (3), (5), and (6) are estimates of a similar form to Equation 1; columns (2) and (4) are estimates of a similar form to Equation 2. The difference is that this uses data for multiple consumer-years and therefore includes additional fixed effects for year and also controls for credit score. *Immigrant* is an indicator for whether the consumer’s SSN was assigned at age 21 or older; *SSN Age* is the consumer’s age at SSN assignment (pooling consumers with SSN Age 20 or younger into one group assigned a value of 20). The outcomes in columns (1) and (2) are whether a consumer has any auto loan inquiry, and columns (3) and (4) are whether a consumer has any mortgage inquiry. The outcome in column (5) is whether a consumer has been rejected for any auto loan, and in columns (6) is whether a consumer has been rejected for any mortgage. Rejections are proxied by inquiry-to-origination non-conversion following the approach in Blattner and Nelson (2024), when an inquiry of a particular credit type (auto loan / mortgage) is not followed by a consumer opening a new account of that same type within 270 days. These rejection estimates condition on the subsample of consumer-years where a consumer makes any inquiry of that credit type in that year. All regressions include fixed effects for birth year and also for a consumer’s first observed ZIP code (First Zip3). All regressions also include fixed effects for each credit score group (no credit score, subprime, nearprime, prime, prime plus, and superprime), and also a linear control for credit score (no scores are given an arbitrary value to prevent these being dropped). Credit scores are measured in the year prior to the inquiry. The units are percentage points. Standard errors are clustered by birth year $\times$ the first two digits of a consumer’s first observed ZIP code. $^+p < .10$; $*p < .05$; $**p < .01$; $***p < .005$.

Dep Var:	Any Auto Loan		Any Mortgage		Any Auto Loan	Any Mortgage
	Inquiry		Inquiry		Rejection	Rejection
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	-1.66*** (0.20)	-1.59*** (0.39)	-0.41** (0.16)	-0.17 (0.29)	3.65*** (0.75)	2.11* (0.92)
SSN Age		-0.02 (0.07)		-0.05 (0.06)		
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip3	X	X	X	X	X	X
F.E. Year	X	X	X	X	X	X
Credit Score Controls	X	X	X	X	X	X
R^2	0.025	0.025	0.017	0.017	0.124	0.158
N	862,015	862,015	862,015	862,015	107,711	53,723
Mean, SSN Age <21	12.71	12.71	6.23	6.23	38.78	55.22

SUPPLEMENTAL APPENDIX: IMMIGRATION AND CREDIT IN THE UNITED STATES

by J. Anthony Cookson, Benedict Guttman-Kenney and William Mullins

A. SUPPLEMENTAL DETAILS ON DATA

A.1 ADDITIONAL DETAILS ON CREDIT REPORTING DATA CONSTRUCTION

We start constructing our data by taking all consumers observed between July 2000 and September 2023 in the University of Chicago Booth TransUnion Consumer Credit Panel, “BTCCP” (TransUnion, 2026). We first queried the BTCCP for this project in September 2023, however, because the data are regularly updated, we observe credit outcomes through to December 2024. We focus the sample on consumers who had at least one tradeline at any point between July 2000 and September 2023. This restriction drops fragmented credit reports, such as those with only credit inquiries. We retain consumers with valid birth dates between 1951 and 2004 because we need birth dates to calculate the year of immigration. As discussed in Gibbs et al. (2025), it is well-established that “consumers” without birth dates are more likely to be credit reports in which debts are fragmented across multiple consumer identifiers; Gibbs et al. (2025) recommends dropping these observations. We start in 1951 for two reasons. First, prior to 1951, there are spikes around particular birth dates, suggesting that some older birth dates may be less accurate in this period. Second, the timing of deaths is not well measured in credit reporting data (Gibbs et al., 2025). Focusing the sample on relatively recent birth years limits this issue. These restrictions produce an anonymous sample of 31.87 million consumers, representative of 318.7 million consumers.

To estimate the year of SSN assignment, *SSN Year*, we construct a lookup table that maps the first five digits of the SSN into the first year of SSN assignment, *SSN Year*, for all SSNs assigned before 2012. We take as the year of SSN assignment the first year the 5-digit sequence was assigned to deal with cases where a 5-digit block spans two or more years.¹ We sent TransUnion the list of all 31.87 million consumers in the full sample, together with this lookup table. TransUnion then matched the consumer list and the lookup table to their underlying data, which includes consumers’ SSNs, returning a dataset with the year of SSN assignment, an indicator for whether a consumer

¹This limitation would also apply to other work that relies on SSN Age (e.g. Bernstein et al., 2025). Our approach potentially introduces some measurement error which would be expected to attenuate cohort comparisons. The fact that we observe sharp differences by SSN Age and also persistent differences in our paired-cohort design suggests that this measurement error does not drive our results.

had any SSN in their data, and the BTCCP consumer identifier. This procedure guarantees that we never observe nor can we infer SSNs for any consumers in the BTCCP.²

Panel A of Table A1 describes how we arrive at our sample. After removing the 21% of consumer identifiers without SSN information, we have 25.2 million consumers. Requiring SSN information appears to remove fragmented credit records with younger ages and credit reports that do not persist over time. It is common to focus research on credit reports with SSNs, most notably, the widely-used Federal Reserve Bank of New York’s Consumer Credit Panel *only* contains information on consumers with SSNs (Gibbs et al., 2025). More fundamentally, our immigrant classification relies on SSN information in a consumer’s TransUnion file. Without a SSN, we cannot establish a consumer’s year of SSN assignment. Fragmentation will be less common in files with SSNs and birth dates because these identifiers are used to consolidate files. In our sample, each SSN is unique; thus, cases with fragmented files may slightly undercount the debt of those consumers.

Without further refinement, this procedure is noisy because all nine digits of SSNs were randomly assigned after mid-2011. Some of these later-assigned SSNs fall into blocks in our lookup table, erroneously assigning them to earlier SSN Years. To limit such misclassification, we remove any consumers who have $SSNAge < 0$, and any consumers with birth years of 1988 or later, as these younger cohorts face significantly higher exposure to post-2011 randomized SSN numbering and do not reach age 37 by the end of our sample. We also exclude consumers if their credit report was established before age 15, indicating a lower quality birth date.

Panel A of Table A1 shows that the resulting “*Clean Sample*” has 19.02 million consumers, 2.23 million of whom we classify as immigrants, based on a SSN Age of 21+. The 11.7 percent of immigrants in this clean sample approximates the 10.2 percent of consumers in the ACS, applying the same birth year restrictions and also only classifying immigrants if their year of immigration was when they are aged 21 or older.

We would not expect exact alignment between our data and the ACS. This is because the ACS estimates populations in America at a point-in-time, whereas our BTCCP data includes consumers who have been present in America up to a point-in-time, which therefore includes consumers who have left the country. If immigrants emigrate more than non-immigrants, this would mean that our share of immigrants in our data would be expected to be higher than in the ACS. A separate

²This classification depends on knowledge of the first five digits of a consumer’s SSN. Bernstein et al. (2025) use Infutor data, which was originally built from consumer credit reporting data. However, since the 1999 Gramm-Leach-Bliley Act, the bureaus cannot sell address lists to third parties, forcing Infutor and its competitors to rely on other data sources.

reason for lower immigration rates in our data is because we only capture legal immigrants with SSNs, whereas the ACS estimates also include illegal immigrants and legal immigrants without SSNs. It is also possible that immigration rates may be higher in our data given that immigrants are under-reported in the ACS, as respondents may not want to disclose their immigrant status.³

We then apply additional restrictions to arrive at the “*Entrant Sample*” used for our paper’s analysis. These restrictions are that consumers have a SSN Age of < 30 and birth years from 1975 to 1987 for the reasons described in the main paper. Within this sample, there are 102 immigration combinations of birth years and SSN Ages. As shown by the counts of consumers by SSN Age in panel B of Appendix Table A1, each SSN Age group in this entrant sample has many consumers, ranging from 57,114 at SSN Age 21 to 22,399 at SSN Age 29. Panel B of Appendix Table A1 also shows that, of the consumers with birth years 1975 to 1987, there are 55,704 consumers assigned a SSN Age of 30+, who we do not include in the Entrant Sample. This means that our SSN Ages 21 to 29 cover 87% of legal immigrants who arrive during adulthood in our data for those birth years.

Further validation shows a close correspondence between our estimates and the ACS estimates of immigrants. In particular, panel A of Appendix Figure A2 shows how the distribution of immigrants by birth year compares in our classification, using SSN Age of 21+ in the BTCCP, to the ACS, using age of immigration of 21+. We see that from birth years 1970 onwards – we use cohorts born in 1975 and onwards in our analysis – our estimate for the percentage of immigrants within each birth year closely aligns with the ACS estimates.⁴ In panel B of Appendix Figure A2, we see that using the Age at SSN Assignment in our data gives a similar distribution of adult age of immigration to the distribution found in the ACS. Both panels suggest that our immigrant classification matches the characteristics of the overall immigrant population *despite* focusing on consumers who have SSNs and whose credit reports are not fragmented. Finally, in panel C of Appendix Figure A2 we show that the immigrant classification delivers a close match to the distribution of immigrants across states found in the ACS. Finally, Appendix Figure A5 shows that the consumers we classify as immigrants are more geographically mobile than non-immigrants, as would be expected with reliable classification. These validation exercises, together with similar validations in the existing literature using the age of SSN assignment (Bernstein et al., 2025, Klopfer and Miller, 2024), suggest that our strategy to classify immigrants is reliable.

³<https://www.census.gov/library/working-papers/2023/adrm/CES-WP-23-42.html>

⁴Before birth year 1969 our classification appears to overestimate the share of immigrants. One reason for this is non-immigrants obtaining SSNs after age 21, but this practice was reduced by the Social Security Amendments of 1972 (P.L. 92-603), which authorized the provision of SSNs to children at the time they first entered school at around age 6 (<https://www.ssa.gov/history/ssn/ssnchron.html>) reducing mis-classification for cohorts born in 1967 onwards.

In an earlier version of our paper, we found our results to be robust to including an additional sample restriction that consumers have a credit report by 2011, which excludes an additional 3% of consumers. The purpose of such a restriction is to focus on consumers most likely to have had their SSNs assigned under the (2011 or earlier) sequential approach that enables us to infer their year of SSN assignment and to substantially limit exposure to those who would have their SSNs assigned using the post-2011 randomized numbering approach. With this restriction, the positive selection of immigrants remains, we estimate that immigrants have significantly higher mean credit score at ages 30 and 40 by 27.0 (compared to 26.7 in this paper without the exclusion) and 26.7 (compared to 24.6 without the exclusion) points, respectively, compared to non-immigrants after controlling for birth year and first Zip5 fixed effects. The estimates for credit access are also similar. For example, by age 37, immigrants are 12.99 (compared to 15.66) percentage points significantly less likely to have ever held an auto loan and 7.54 (compared to 8.91) percentage points significantly less likely to have held a mortgage, compared to non-immigrants after controlling for birth year and first ZIP5 fixed effects.

A.2 ADDITIONAL DETAILS ON THE SURVEY OF CONSUMER FINANCES (2022)

We use the 2022 wave of the SCF as this is the only wave that includes information on whether a consumer is an immigrant and the number of years since immigration. When using the SCF, we apply similar birth year restrictions to our credit reporting data, and doing so reduces the SCF sample size from 4,595 to 1,813 respondents, of whom we classify 213 as immigrants based on an age of immigration of 21+. We study respondents in their twenties, thirties, and forties in the 2022 survey year (corresponding to birth years 1972 to 2000). We use this restriction because consumers in their fifties and older are expected to have limited credit demand irrespective of immigration status or credit supply. Given the small survey sample size, we retain respondents with ages of immigration older than 30, in contrast to the restriction used in the credit reporting data and the ACS.

When analyzing the SCF, we use cross-sectional regressions of the same overall form as Equations 1 and 2, with a few minor changes. Rather than using *SSN Age*, we directly observe the age of immigration in the survey, and therefore $\mathbb{I}\{Immigrant\}_i$ measures whether a respondent immigrated at age 21 or later in life, and instead of *SSN Age_i* we have *Immigrant Age_i*, a variable for each year of immigrating at age 21 or later in life. Given the small survey sample size, we increase the precision of our regression estimates by including demographic controls for normal in-

come (winsorized at the 99th percentile) and fixed effects for birth year ($\mu_{b(i)}$), male, race, hispanic, spouse, marital status, household size, number of children, and education levels, but cannot include geographic fixed effects ($\eta_{g(i)}$) as these are not in the data. The SCF is conditional on a consumer being in the United States when surveyed in 2022 and, therefore our results are conditional on not emigrating. We use the SCF’s recommended survey weights in our regressions to correct for the over-sampling of high-wealth consumers in the survey’s design, and we use heteroskedasticity-robust standard errors and follow the SCF’s recommended multiply-imputed variance estimation technique to account for missing data.

In all of the regression in Appendix Tables A9 and A10, the age of immigration is constructed from the survey variable x6906 (years lived in the United States) and from variable x8022 (age). The indicator for immigration status (*Immigrant*) takes a value of one if the consumer’s age of immigration was at 21 or older, and zero otherwise. The variable *Immigration Age*, the consumer’s age at immigration, pools all consumers with ages of immigration at 20 or younger at a value of 20.

In Appendix Table A9, the outcome in columns 3 to 6 of Panel A is whether a consumer has any outstanding auto loan debt or lease on any of their vehicles (constructed from the auto financing and leasing variables x2102, x2208, x2308, x2408, x7157). The mean of this variable may appear to be lower than one might expect, given that most car purchases in the United States are financed with debt (see cites in Gibbs et al., 2025), however, this makes sense because some consumers will have previously taken out auto loans but have paid this debt off by the time of being surveyed.

General debt attitudes, the outcome in columns (7) and (8) of Appendix Table A9 Panel A are measured by the survey questions that state: “Now I would like to ask you some questions about how you feel about credit. In general, do you think it is a good idea or a bad idea for people to buy things by borrowing or on credit?” (constructed from variable x401), with the variable having three potential values: 1 if they regard it as a good idea, 0 if good in some ways, bad in others, and -1 if they regard it as a bad idea. In this table we multiply estimates by 100 to be on a similar scale to other estimates. In columns (1) to (4) of Panel A of Appendix Table A10, we decompose this variable into binary measures. In columns 1 and 2, this outcome takes a value of 1 if they regard it as a good idea, and zero otherwise. In columns 3 and 4, this outcome takes a value of 1 if they regard it as a bad idea, and zero otherwise.

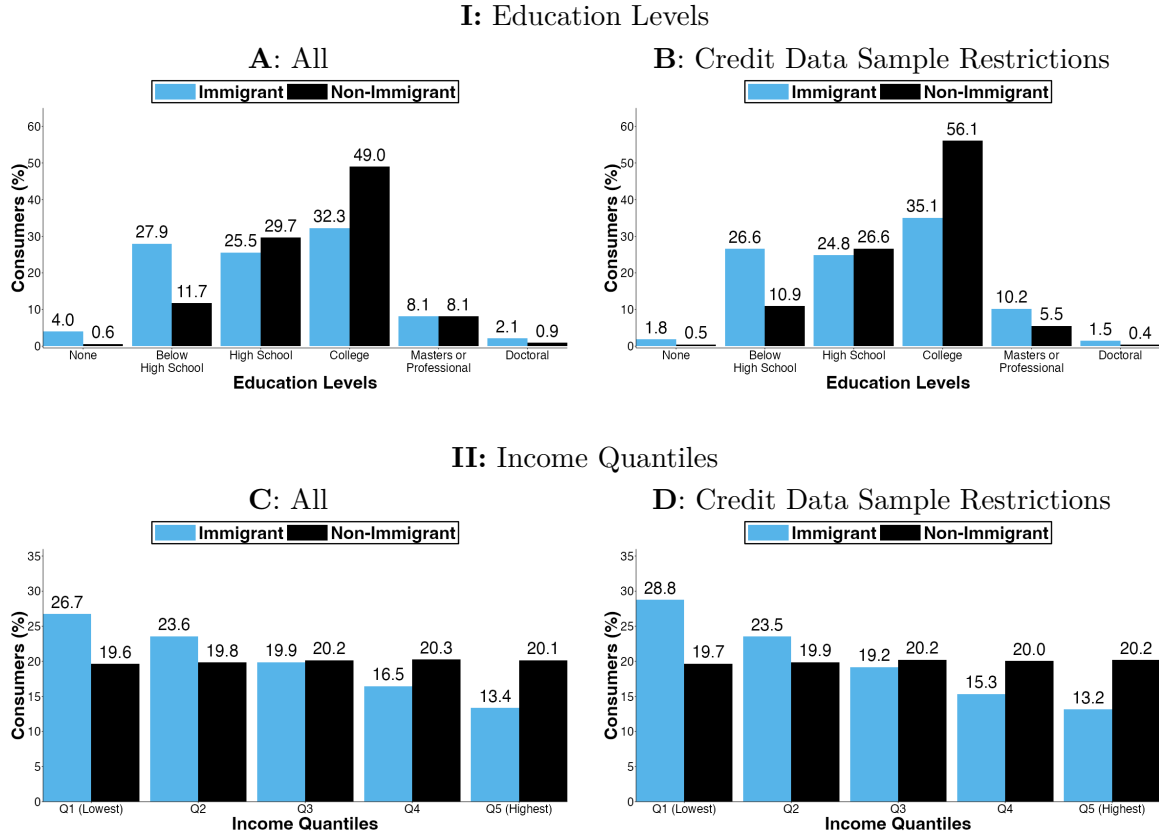
The outcome (constructed from variable x405) in columns (1) and (2) of Appendix Table A9, and also used as a control variable *Auto Debt Attitude* in columns (5) and (6) of this same figure, is

a binary variable for whether they feel it is all right to borrow money to finance the purchase of a car. This is following the prompt “people have many different reasons for borrowing money which they pay back over a period of time, please tell me whether you feel it is all right for someone like yourself to borrow money...”, taking a value of 1 if they respond yes and a value of 0 if they respond no.

The outcomes in columns (5) to (10) of Appendix Table A10 Panel A are all binary variables responding to different scenarios following the prompt “people have many different reasons for borrowing money which they pay back over a period of time. For each of the reasons I read, please tell me whether you feel it is all right for someone like yourself to borrow money...”, taking a value of 1 if they respond yes and a value of 0 if they respond no. The outcome in columns 5-6 (constructed from variable x403) is whether they feel it is all right for someone like yourself to borrow money to cover living expenses when income is cut. The outcome in columns 7-8 (constructed from variable x402) is whether they feel it is all right for someone like yourself to borrow money to cover the expenses of a vacation trip. The outcome in columns 9-10 (constructed from variable x406) is whether they feel it is all right for someone like yourself to borrow money to finance educational expenses.

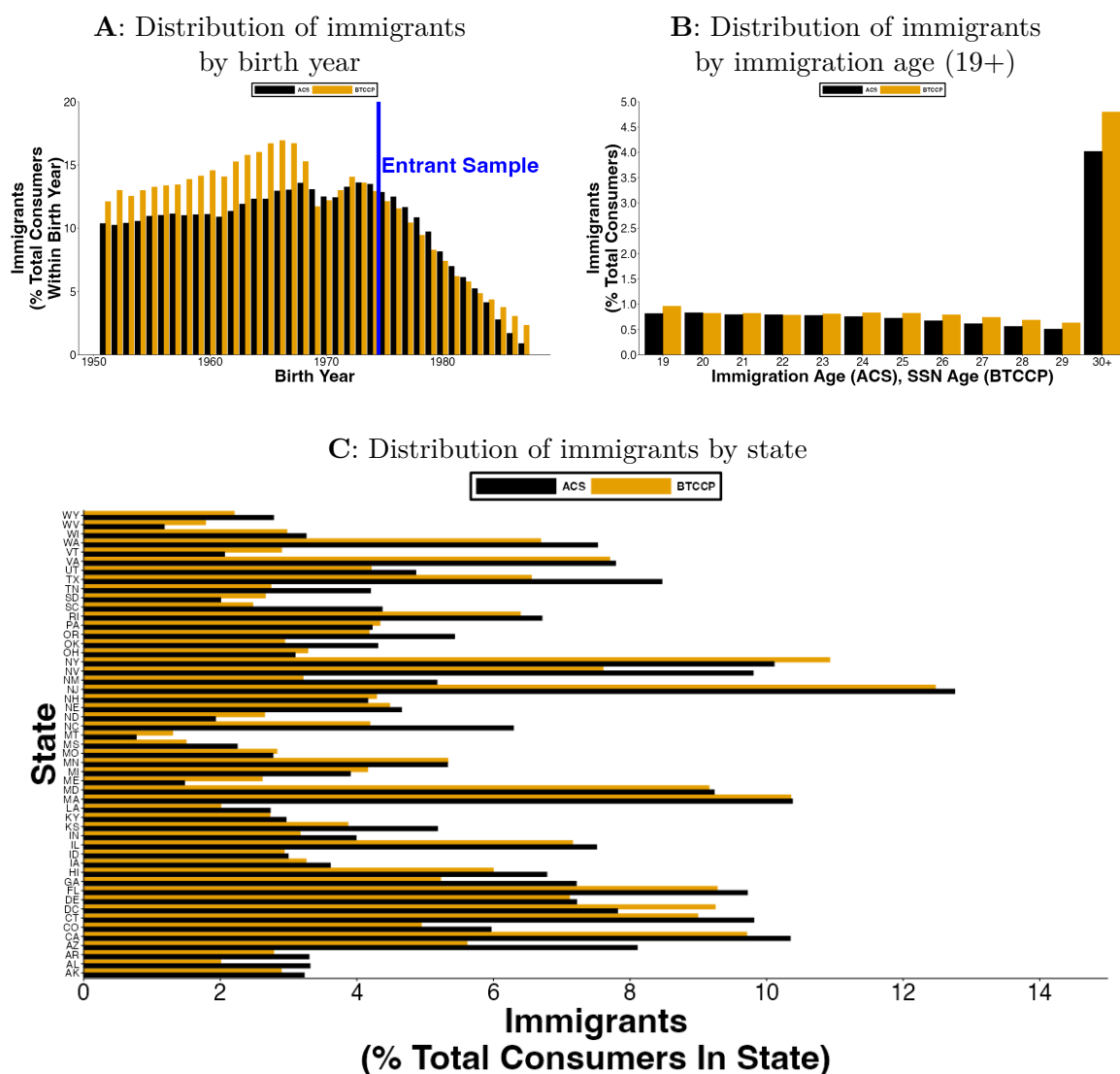
B. SUPPLEMENTAL FIGURES AND TABLES

Figure A1: Distribution of education and income of immigrants compared to non-immigrants in the American Community Survey (ACS)



This figure summarizes data computed from the 5-year 2010 American Community Survey (ACS). Panels A and C includes all consumers with positive income and who are aged between 18 and 80. Panels B and D also restrict to birth years 1975 to 1987 and excludes immigrants who arrived from age 30 or older to create a sample that is comparable to the Entrant sample that we use in our analysis of credit data. In all panels, immigrants are classified as those whose age of immigration is at age 21 or older, with everyone else classified as non-immigrants. In each panel, each black bar shows the baseline means for the non-immigrant group. In each panel, each blue bar shows the baseline mean added to the estimates from regressions on an indicator for immigration age, where the regressions also control for birth year, geographic area (Consistent Public Use Microdata Areas), and survey year. The outcomes in Panels A and B are indicators for each level of education. The outcomes in Panels C and D are indicators for each quintile of household income.

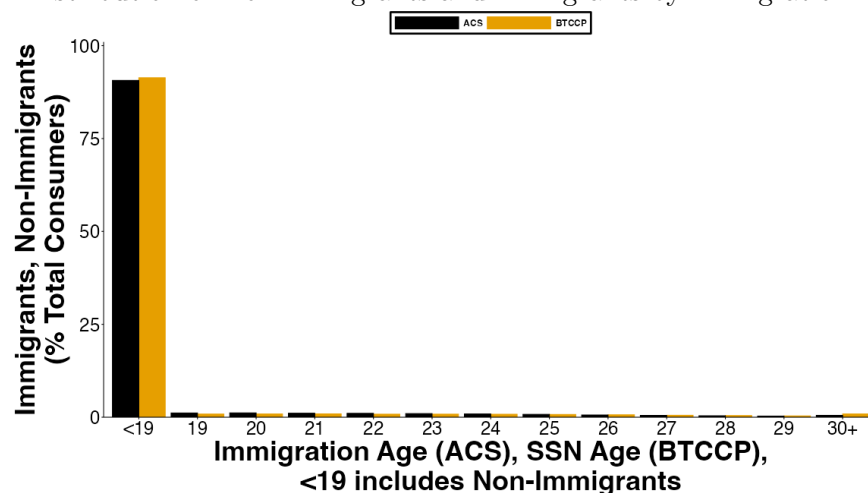
Figure A2: Immigrant classification in our data (BTCCP) versus the American Community Survey (ACS)



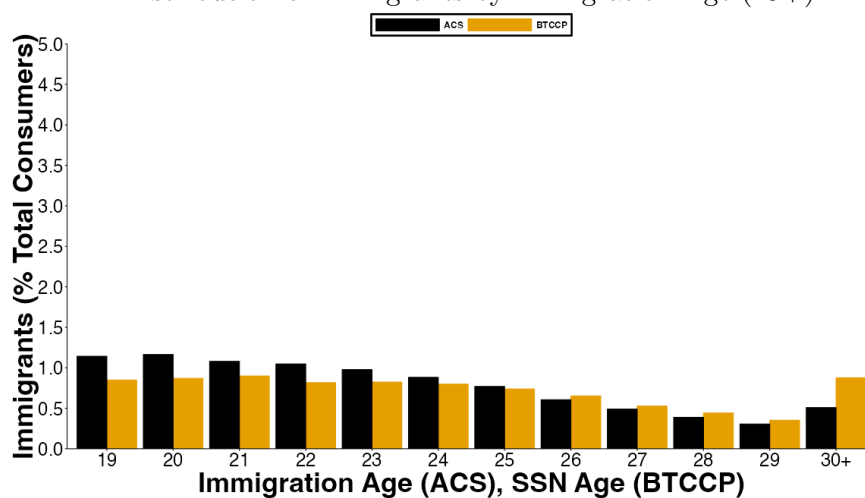
This figure compares our classification of immigrants in the Booth TransUnion Consumer Credit Panel (BTCCP) to comparable statistics computed from the American Community Survey (ACS). Panel A presents the share of immigrants in each birth year in the ACS (black) versus BTCCP (yellow). In Panel A, immigrants are defined as SSN at age 21+ in the BTCCP and if their age of immigration is 21+ in the ACS. The birth years to the right of the blue vertical line are the “Entrant Sample” of birth years 1975 to 1987 that are used for our analysis. Panel B presents the share of each sample by age at immigration, non-immigrants or those that immigrate at age 18 or younger are included in the denominator but the bar is excluded from this figure to ease presentation, it is 88.15% for the ACS and 86.52% for the BTCCP. Panel C shows the shares of immigrants in each state as measured in the ACS and BTCCP.

Figure A3: Immigrant classification in our data (BTCCP) versus the American Community Survey (ACS) for entrant sample, with SSN Age 30+ added

A: Distribution of Non-Immigrants and Immigrants by Immigration Age

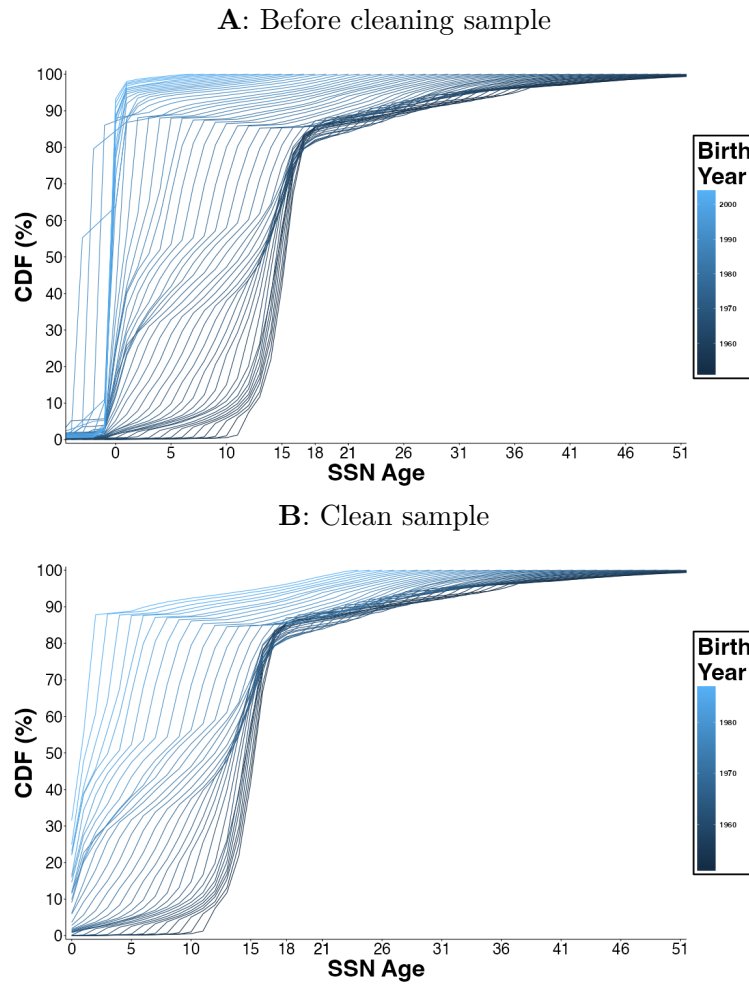


B: Distribution of Immigrants by Immigration Age (19+)



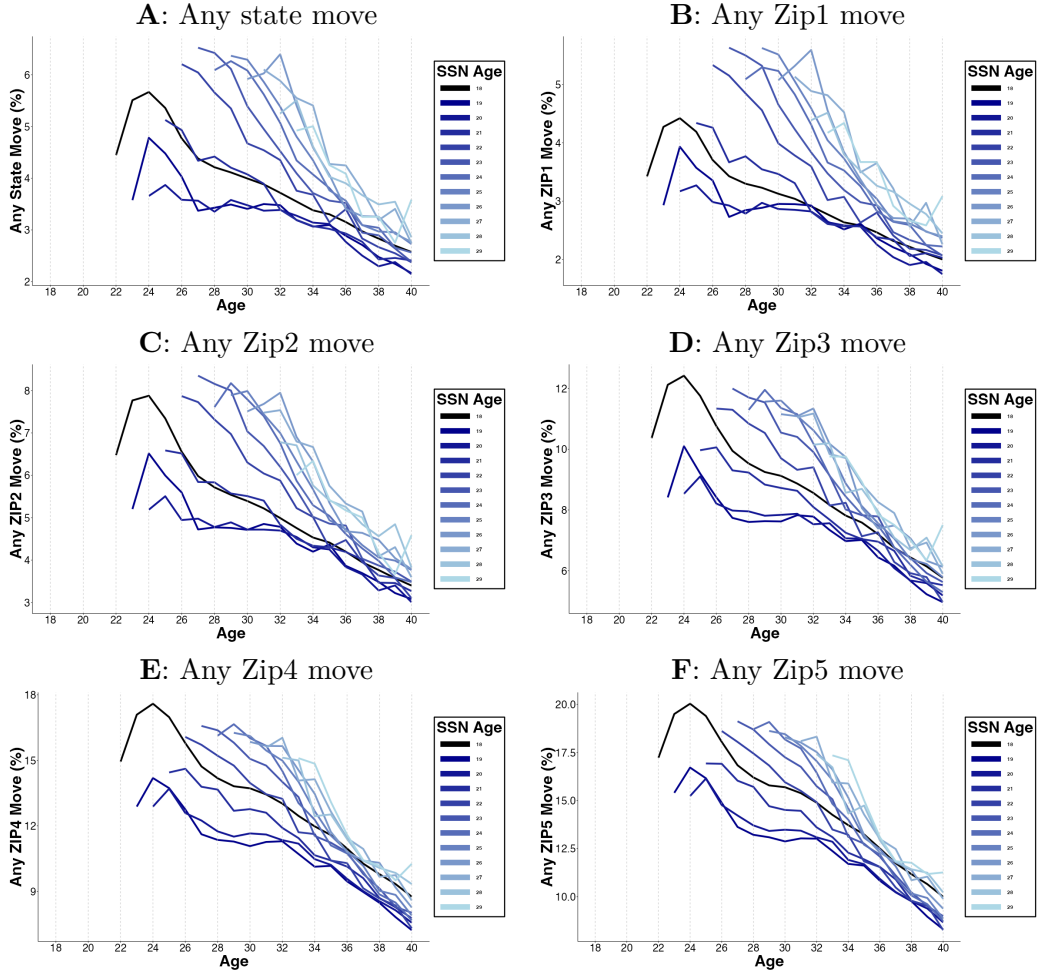
This figure compares our classification of immigrants in the BTCCP Entrant Sample to comparable statistics computed from the American Community Survey (ACS). The figure shows bars for immigrants with Age of Immigration (ACS) / SSN Age of 30+, these are consumers who are not included in the Entrant Sample used for analysis. Panel A presents the share of immigrants by immigration age, where < 19 includes non-immigrants or those that immigrate at age 18 or younger. Panel B presents a zoomed in version of Panel A that only shows the bars for the subset of immigration ages from 19 to 30+.

Figure A4: Distributions of SSN Age by birth year



These panels show the CDFs of SSN Age for each birth year. Panel A shows data before cleaning (only removing consumers without SSNs). Panel B restricts to consumers after cleaning the data. These patterns of SSN Ages by birth years are consistent with Klopfer and Miller (2024) using administrative Social Security Administration data.

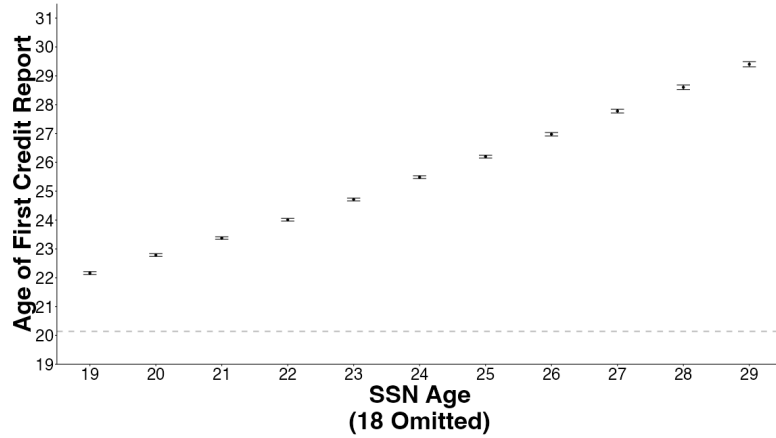
Figure A5: Lifecycle of geographic mobility by SSN Age cohorts



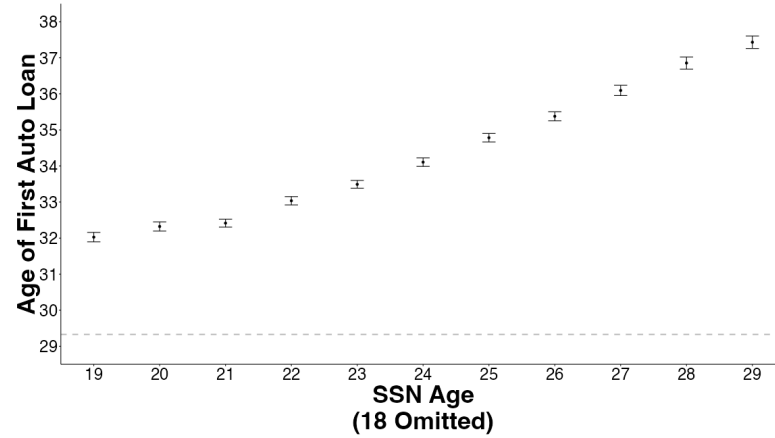
For each SSN Age cohort, this figure shows the share of consumers at each age who move state (Panel A), ZIP1 (Panel B), ZIP2 (Panel C), ZIP3 (Panel D), ZIP4 (Panel E), and ZIP5 (Panel F). The black line (SSN Age 18) pools all consumers with SSN Age 18 or younger. Lighter colors indicate later SSN Age cohorts. $Age = SSN\ Age$ is indicated by the circles on each line. This figure uses data for birth years from 1982 to 1987. Consumers with the birth years 1985, 1986, and 1987, are not observed for ages 40, 39 to 40, and 38 to 40, respectively, by the end of our data in 2024.

Figure A6: Age at first credit product by SSN Age

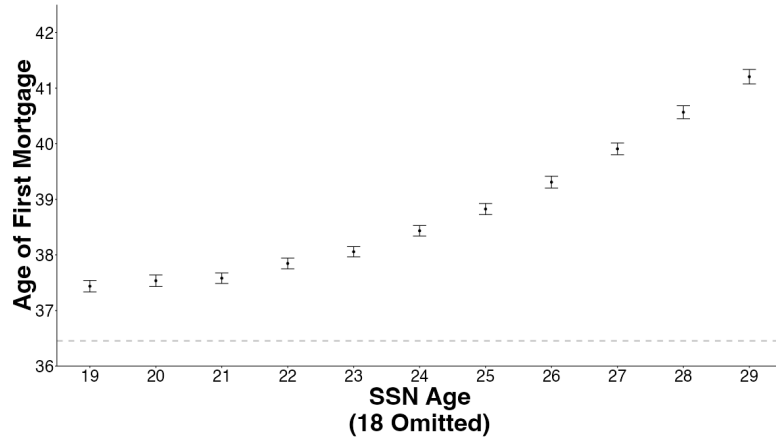
A. Age at first credit report



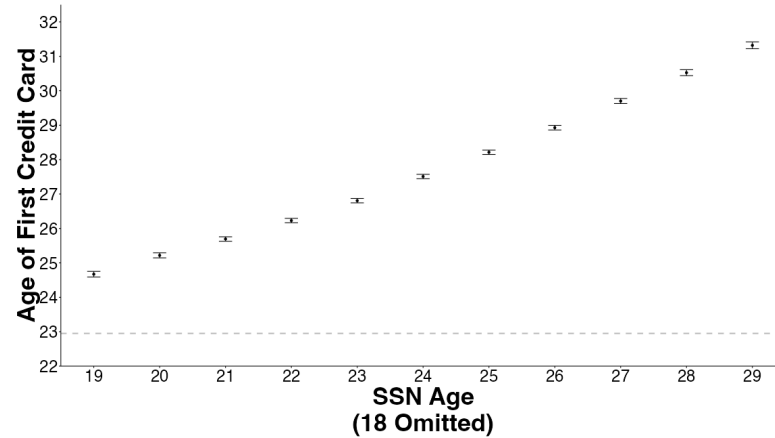
B. Age at first auto loan



C. Age at first mortgage

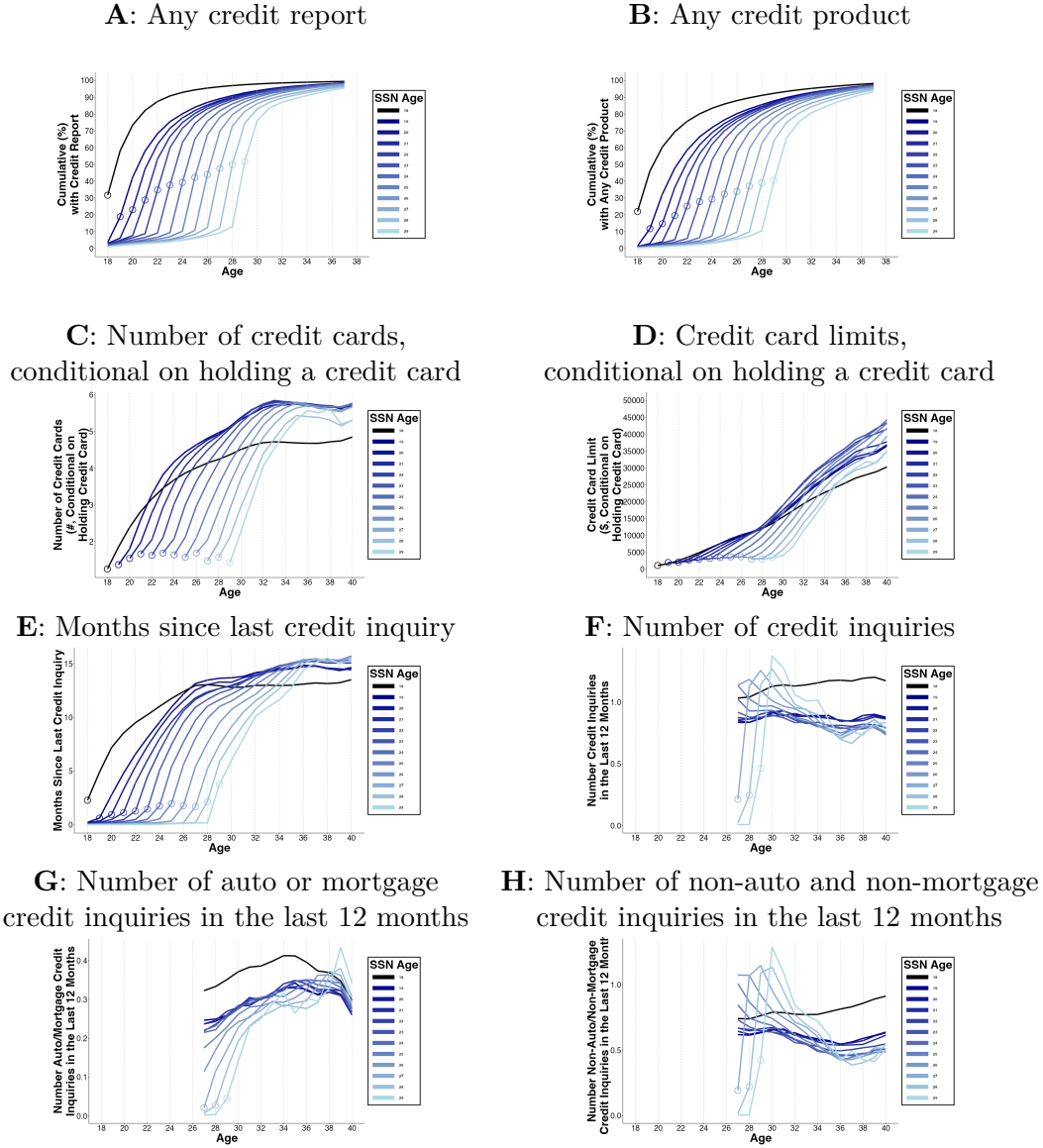


D. Age at first credit card



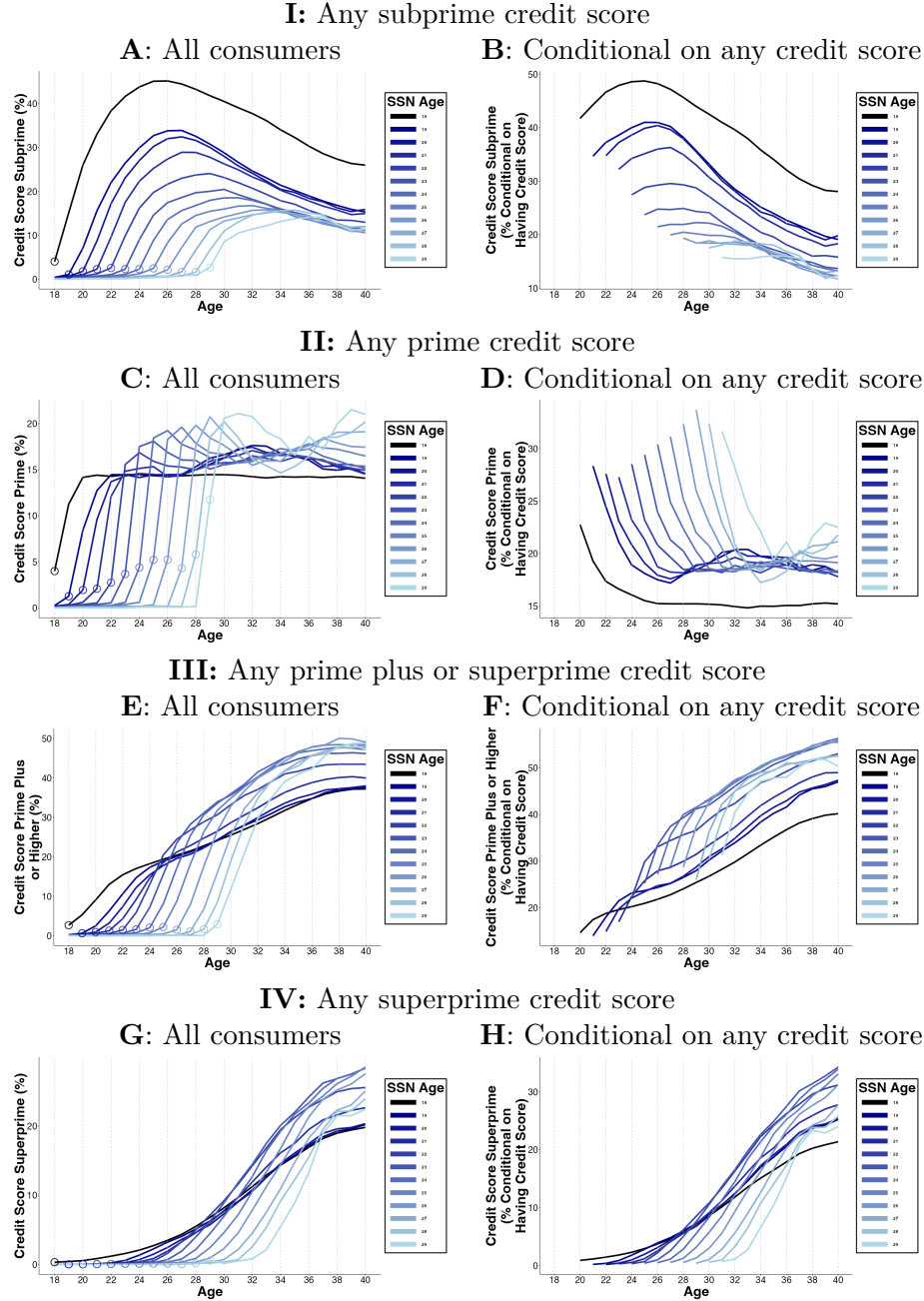
This figure presents estimates and 95% confidence intervals for age at first credit report (Panel A), age at first auto loan (Panel B), age at first mortgage (Panel C), and age at first credit card (Panel D) separately by SSN Age cohorts. The estimates are constructed from an individual-level regression of each outcome on SSN Age indicators, Birth Year fixed effects, and fixed effects for consumers' first observed ZIP code. The baseline mean for the omitted category (SSN Age 18 or lower) is indicated by the dashed gray line. In these regressions, for consumers that do not access a product by the end of our data, we assign them a value that corresponds to accessing that product in 2025, one year after the end of our data. Standard errors are clustered by birth year $\times$ the first two digits of a consumer's first observed ZIP code.

Figure A7: Additional credit outcomes by SSN Age cohorts



For each SSN Age cohort, this figure presents the evolution of a consumer having any credit report (Panel A) and any credit product (Panel B) by age, using data for birth years from 1975 to 1987. Because Panels A and B are cumulative charts we end them at age 37, whereas for the other panels the estimates for ages 38 to 40 account for this attrition. Consumers with the birth years 1985, 1986, and 1987, are not observed for age 40, ages 39 to 40, and ages 38 to 40, respectively by the end of our data in 2024. Panels C to H use data for birth years from 1982 to 1987 to show the mean number of credit cards, conditional on holding a card at that age (Panel C), the mean credit card limits, conditional on holding a credit card at that age (Panel D), and the mean number of months since last credit inquiry (Panel E) at each age for consumers of each SSN Age. The outcomes in Panels F, G, and H are the total number of credit inquiries in the last twelve months (Panel F), the subset that are for auto or mortgages (Panel G), and the difference between these variables to show the remainder, i.e., inquiries that are not for auto loans or mortgage loans (Panel H), respectively. These last three variables are only observed for the years 2009 to 2024 and the data does not allow us to further differentiate credit types applied for. In all these panels, the black line (SSN Age 18) pools all consumers with SSN Age 18 or younger. Lighter colors indicate later SSN Age cohorts. $Age = SSN\ Age$ is indicated by the circles on each line.

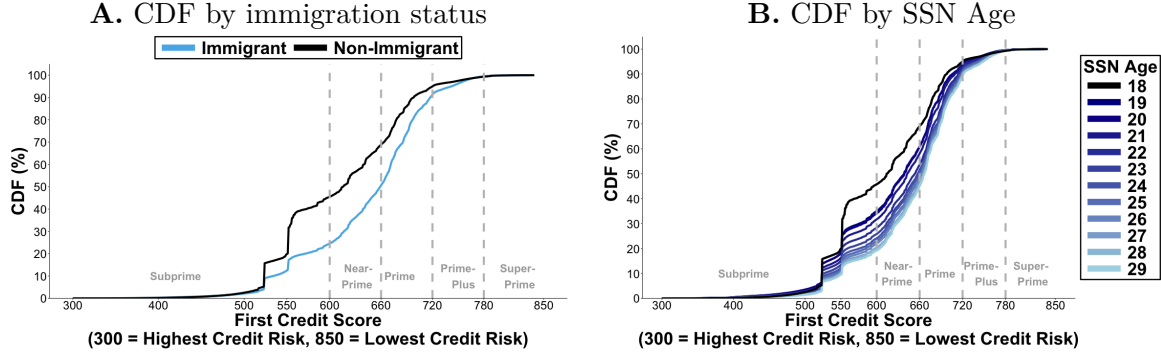
Figure A8: Lifecycle of credit scores by SSN Age cohorts



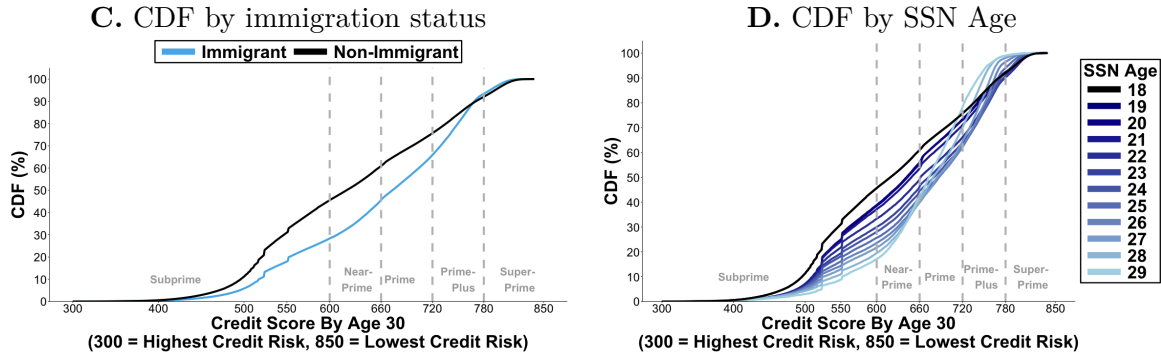
For each SSN Age cohort, this figure presents the evolution of credit scores by age. In Panels A and B, Subprime Credit Score is a credit score is ≤ 600 . In Panels C and D, Prime Credit Score is a credit score of 661 to 720. In Panels E and F, Prime Plus or Superprime Credit Score is credit score of 721 or higher. In Panels G and H, Superprime Credit Score is credit score of 781 or higher. The black line (SSN Age 18) pools all consumers with SSN Age 18 or younger. Lighter colors indicate later SSN Age cohorts. $Age = SSN Age$ is indicated by the circles on each line. This uses data for birth years from 1982 to 1987.

Figure A9: Distribution of credit scores

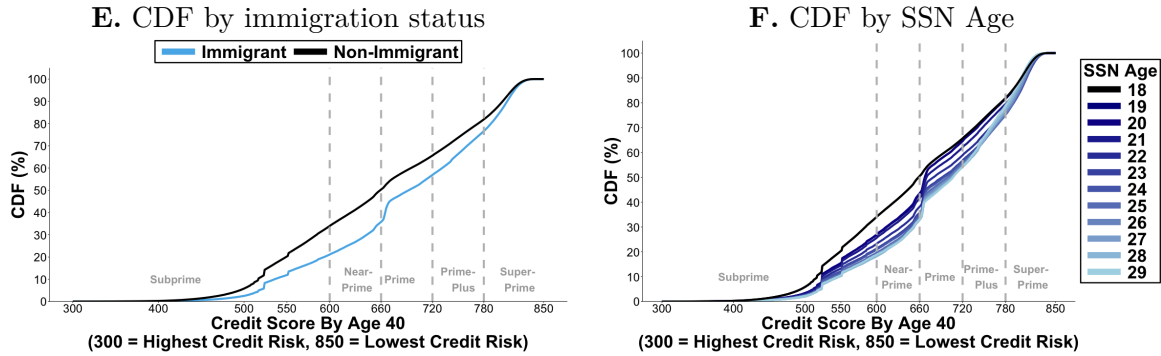
I: First credit score



II: Credit score by age 30

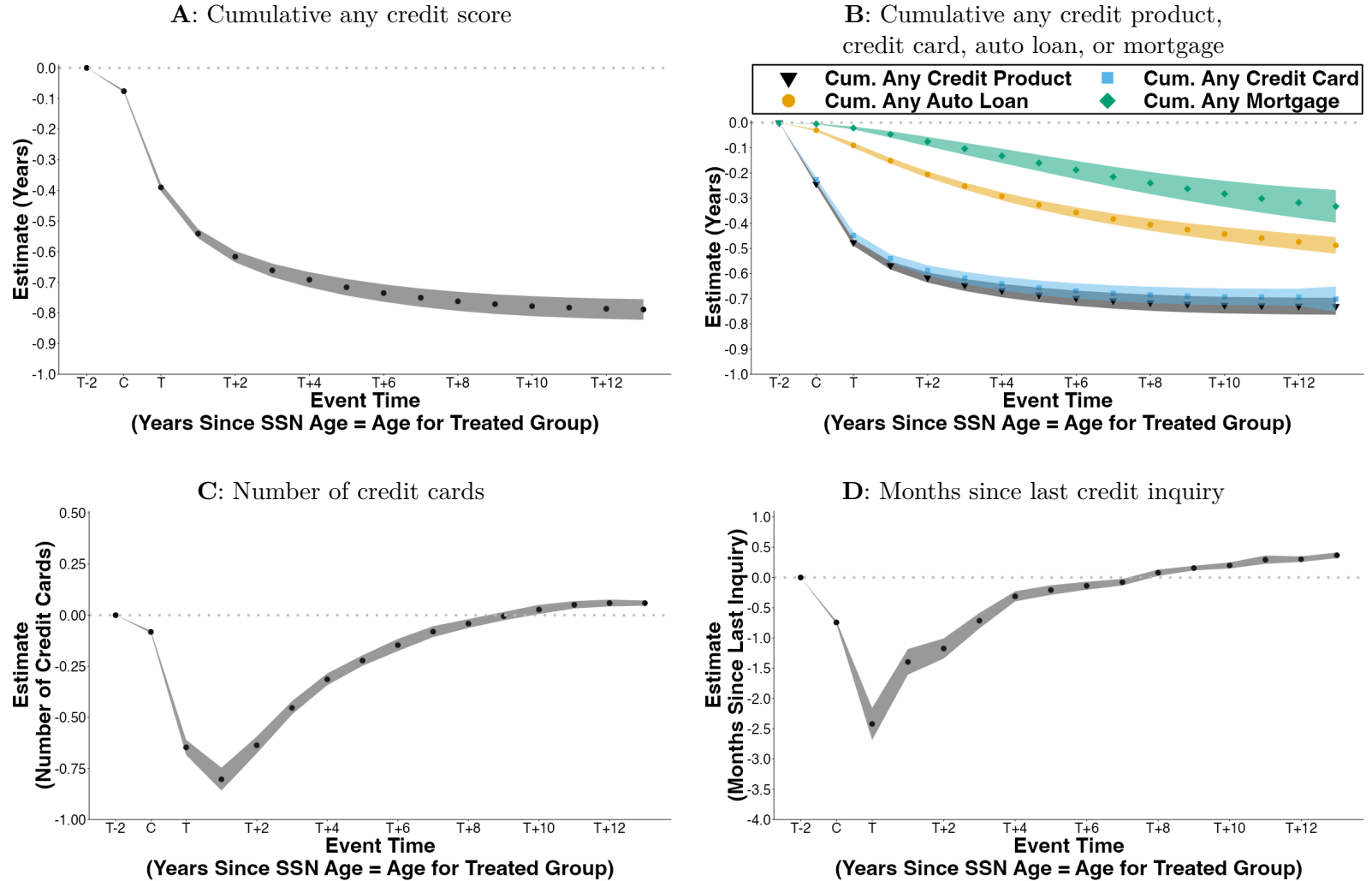


III: Credit score by age 40



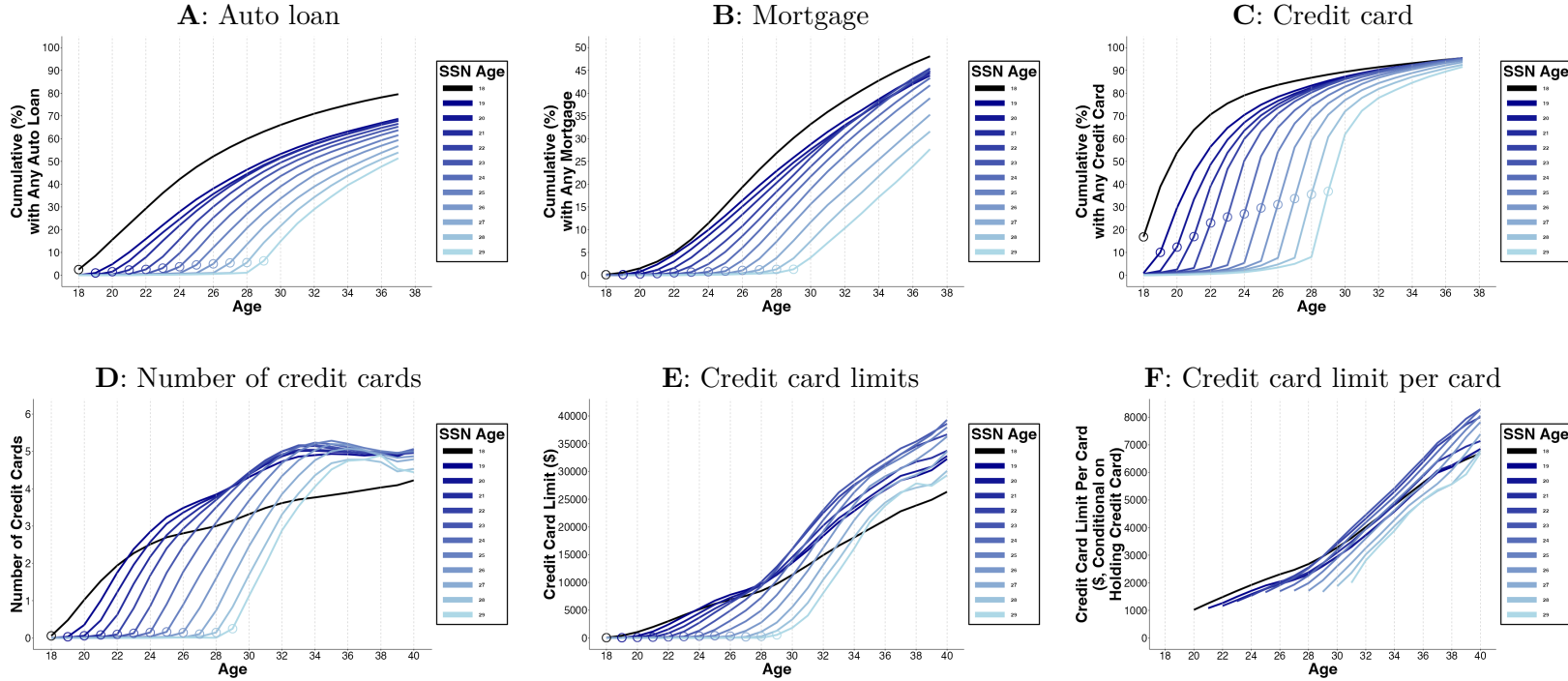
Panels A and B show the CDFs of the first credit score observed for a consumer at any point in our data. Panels C and D show CDFs of credit score by Age 30. Panels E and F show CDFs of credit score by Age 40. Panels A, C, and E show CDFs by immigration status, where immigrants are defined as SSN Age 21 to 29. Panels B, D, and F show CDFs by SSN Ages, where SSN Age 18 includes consumers that are assigned an SSN at Age 18 or younger. The outcomes in Panels C to F are conditional on having a non-missing credit score by ages 30 and 40 respectively (for consumers not observed at a given age, we use the score at the oldest age observed before that age). Panels A and B are also conditional on having a non-missing credit score, and we also drop consumers with credit scores that are first observed in July 2000, where our data begins, as credit scores that are first observed at this point-in-time may not necessarily reflect a consumer's first credit score, just a consumer's first score in our data.

Figure A10: Paired cohorts: dynamics of credit access



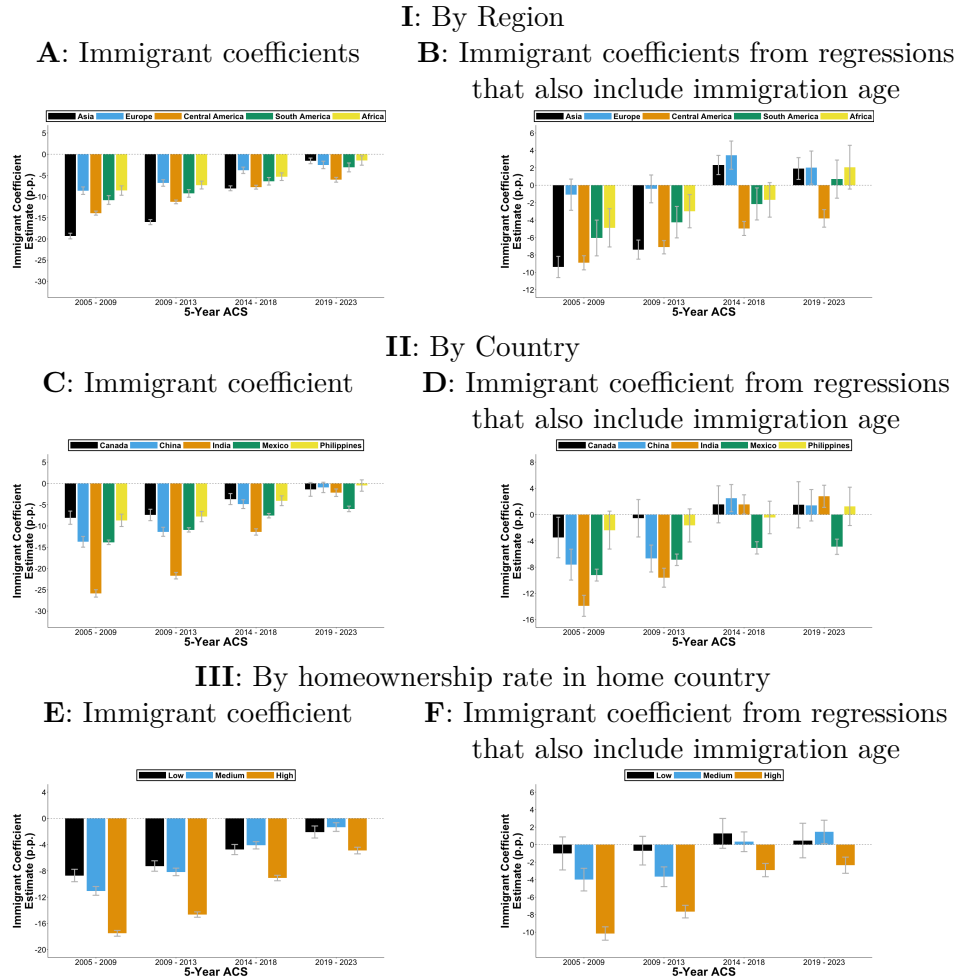
This figure presents dynamic estimates for differences in credit access (of different types) between a cohort with SSN Age = s and a cohort with SSN Age = $s - 1$ matched at the same age. The differences are presented in event time where C is the year when age equals SSN Age for the $s - 1$ cohort and T is the same for the s cohort: cumulative years with any credit score (Panel A), cumulative years with any credit product, any credit card, any auto loan, and any mortgage (Panel B), the number of credit cards (Panel C), and months since last credit inquiry (Panel D). The shaded areas indicate 95% confidence intervals, calculated using wild cluster bootstrapping by birth year using Webb weights and 10,000 iterations.

Figure A11: Lifecycle of credit access by SSN Age cohorts, restricting sample to consumers still observed in 2024



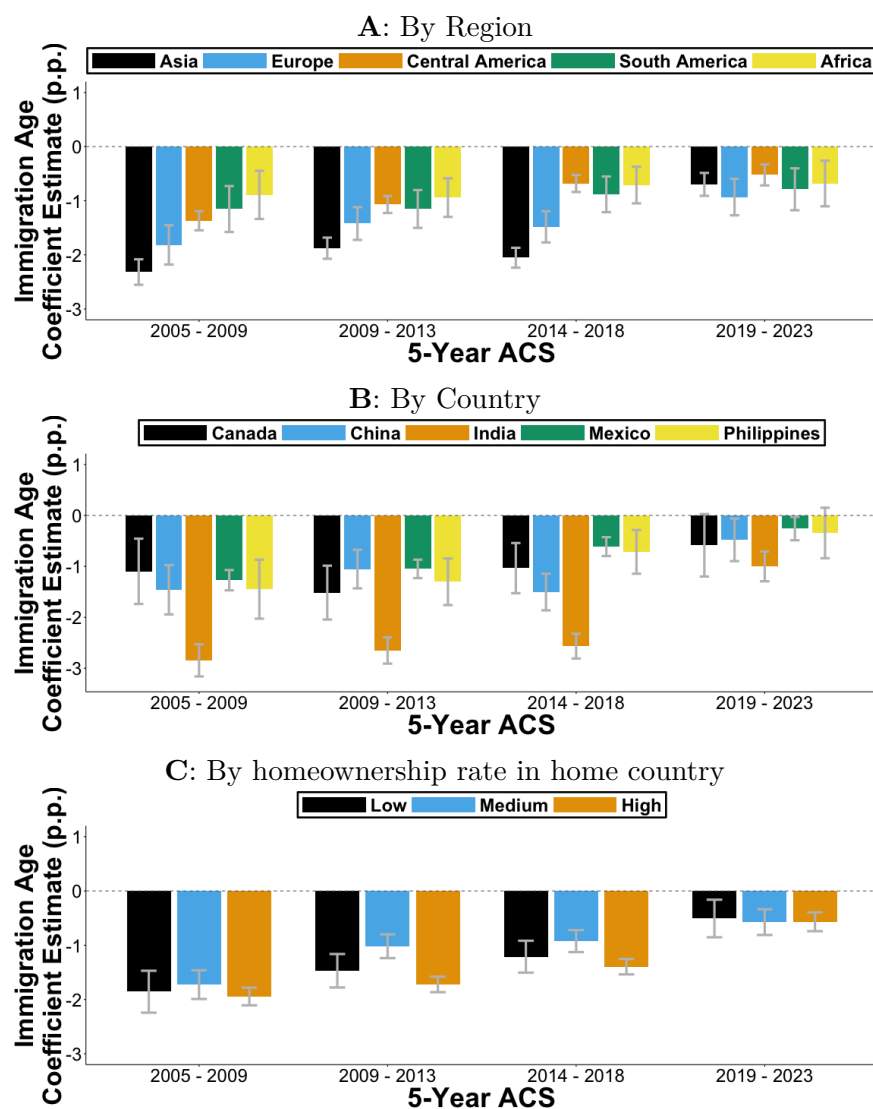
This figure restricts the sample to consumers with a non-missing credit score in 2024. For each SSN Age cohort, this figure shows the cumulative share of consumers at each age who have ever had an auto loan (Panel A), a mortgage (Panel B), or a credit card (Panel C). Panels A, B, and C use data for birth years from 1975 to 1987. Because Panels A, B, and C are cumulative charts we end them at age 37, whereas for the other panels the estimates for ages 38 to 40 account for this attrition. Consumers with the birth years 1985, 1986, and 1987, are not observed for age 40, ages 39 to 40, and ages 38 to 40, respectively, by the end of our data in 2024. Panels D, E, and F use data for birth years from 1982 to 1987 to show the mean number of credit cards (Panel D), the mean credit card limits (Panel E), and the mean credit card limit per card (Panel F) at each age for consumers of each SSN Age. The black line (SSN Age 18) pools all consumers with SSN Age 18 or younger. Lighter colors indicate later SSN Age cohorts. $Age = SSN\ Age$ is indicated by the circles on each line.

Figure A12: American Community Survey: Immigrant coefficient estimates on mortgage use, by heterogeneous immigrant groups



This figure uses public data from the 5-year American Community Survey (ACS) waves 2005-2009, 2009-2013, 2014-2018, and 2019-2023, keeping consumers with birth years 1975 to 1987, non-zero income, and immigrants who immigrated between ages 21 and 29 along with non-immigrants. The outcome in all panels is whether a consumer has any mortgage. Each x-axis category shows results from a different 5-year ACS wave. Each bar shows estimates from a separate cross-sectional regression for heterogeneous groups of consumers: Panels A and B restrict to immigrants based on their geographic region of birth, Panels C and D restrict to immigrants based on their country of birth, and Panels E and F adapts Gorbach and Schubert (2026) to restrict to immigrants based on terciles of the homeownership rates in their country of birth, for 84 countries observed. All regressions include non-immigrants. Each bar shows coefficient estimates on an indicator for immigration status (*Immigrant* is an indicator for whether the consumer's age of immigration was at age 21 or older). The regressions in Panels B, D, and F also include *Immigration Age*, the consumer's age at immigration (pooling consumers with an age of immigration of 20 or younger into one group). The regressions in Panels A, C, and E do not include *Immigration Age*. All regressions include fixed effects for the consumer's birth year, geographic area (Consistent Public Use Microdata Areas), sex, education levels, race, hispanic, survey year, employment status and occupation. The regressions also include a control for Log (Income), the log of household income that is winsorized at 99.9th percentile and is scaled to mean zero and standard deviation one. Units are percentage points. 95% confidence intervals are calculated using heteroskedasticity-robust standard errors.

Figure A13: American Community Survey: Immigration Age coefficient estimates on mortgage use, by heterogeneous immigrant groups



This figure uses public data from the 5-year American Community Survey (ACS) waves 2005-2009, 2009-2013, 2014-2018, and 2019-2023, keeping consumers with birth years 1975 to 1987, non-zero income, and immigrants who immigrated between ages 21 and 29 along with non-immigrants. The outcome in all panels is whether a consumer has any mortgage. Each x-axis category shows results from a different 5-year ACS wave. Each bar shows estimates from a separate cross-sectional regression for heterogeneous groups of consumers: Panel A restricts to immigrants based on their geographic region of birth, Panel B restricts to immigrants based on their country of birth, and Panel C adapts Gorbach and Schubert (2026) to restrict to immigrants based on terciles of the homeownership rates in their country of birth, for 84 countries observed. All regressions include non-immigrants. Each bar shows coefficient estimates for *Immigration Age*, the consumer's age at immigration (pooling consumers with an age of immigration of 20 or younger into one group). All regressions include an indicator, *Immigration*, for whether a consumer's age of immigration was at age 21 or older, along with fixed effects for the consumer's birth year, geographic area (Consistent Public Use Microdata Areas), sex, education levels, race, hispanic, survey year, employment status and occupation. The regressions also include a control for Log (Income), the log of household income that is winsorized at 99.9th percentile and is scaled to mean zero and standard deviation one. Units are percentage points. 95% confidence intervals are calculated using heteroskedasticity-robust standard errors.

Table A1: Data sample

Panel A presents an observation funnel that details how we obtain our final sample starting from the full BTCCP 10% sample. As each sample restriction is applied, the table details how the number of consumers changes, until we obtain the dataset used for our analysis: the *Entrant Sample*. Panel B presents observation counts for the “Clean Sample” (indicated in Table A1) by SSN Age in column 1. Our entrant sample additionally drops the 55,704 consumers who have SSN Age 30+, these consumers are included in Panel B of this table for completeness.

Panel A: Sample construction

TransUnion Consumers	
... born 1951 to 2004	31,869,445
... with a SSN in TransUnion	25,237,917
Clean Sample	19,019,191
... SSN Age <21	16,793,939
... SSN Age 21+	2,225,252 (11.70%)
... Immigrant Cohorts (Birth Year x SSN Year)	775
Entrant Sample (SSN Age < 30)	6,299,287
... SSN Age < 21	5,914,891
... SSN Age 21-29	384,396 (6.10%)
... Immigrant Cohorts (Birth Year x SSN Year)	102

Panel B: Counts of consumers by SSN Age: “clean sample” and “entrant sample”

SSN Age	Clean Sample	Entrant Sample
<19	16,455,525	5,805,684
19	182,481	53,950
20	155,933	55,257
21	155,743	57,114
22	149,210	51,894
23	153,427	52,314
24	157,685	50,705
25	156,152	46,924
26	150,045	41,481
27	140,610	33,525
28	130,243	28,040
29	119,788	22,399
30+	912,349	55,704

Table A2: Credit access, with additional geographic fixed effects

This table replicates Tables 2 and 4 with additional geographic fixed effects. The specification includes baseline fixed effects (birth year of the consumer, and consumers' first observed ZIP code, *First Zip5*), and adds last observed ZIP code (*Last Zip5*), longest observed ZIP code (*Longest Zip5*), and the number of unique ZIP codes observed (*Number Zip5*) as a measure of consumer mobility. The outcomes in Panel A are: age at first credit report (columns 1 and 2), age at first credit product (columns 3 and 4), age at first auto loan (columns 5 and 6), age at first mortgage (columns 7 and 8), and age at first credit card (columns 9 and 10). The outcomes in Panel B are: the probability of having ever held an auto loan by age 37 (columns 1 and 2), the probability of having ever held a mortgage by age 37 (columns 3 and 4), the probability of having ever held a credit card by age 37 (columns 5 and 6). The units in Panel A are years of age and the units in Panel B are percentage points. Standard errors are clustered by birth year $\times$ the first two digits of a consumer's first observed ZIP code. $*p < .05$; $**p < .01$; $***p < .005$.

Panel A: Age of credit products first taken out

Dep Var: Age at First...	Credit Report		Credit Product		Auto Loan		Mortgage		Credit Card	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Immigrant	5.15*** (0.02)	2.07*** (0.02)	4.73*** (0.02)	1.81*** (0.03)	3.88*** (0.04)	1.60*** (0.04)	1.72*** (0.03)	0.03 (0.04)	4.21*** (0.02)	1.49*** (0.03)
SSN Age		0.72*** (0.01)		0.68*** (0.01)		0.53*** (0.01)		0.39*** (0.01)		0.63*** (0.01)
F.E. Birth Year	X	X	X	X	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X	X	X	X	X
R^2	0.260	0.275	0.200	0.207	0.180	0.181	0.158	0.159	0.173	0.176
N	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372
Mean, SSN Age <21	20.19	20.19	21.56	21.56	29.40	29.40	36.49	36.49	22.99	22.99

Panel B: Probability of having held a credit product by age 37

Dep Var: By Age 37, has...	Auto Loan		Mortgage		Credit Card	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	-10.61*** (0.17)	-5.62*** (0.21)	-5.21*** (0.17)	1.39*** (0.21)	0.09 (0.07)	0.37*** (0.10)
SSN Age		-1.16*** (0.04)		-1.54*** (0.04)		-0.06*** (0.02)
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X
R^2	0.137	0.137	0.151	0.151	0.070	0.070
N	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372
Mean, SSN Age <21	76.06	76.06	44.63	44.63	94.12	94.12

Table A3: Creditworthiness and credit card outcomes with additional geographic fixed effects

Panel A replicates Table 3 with additional geographic fixed effects. The specification includes baseline fixed effects (birth year of the consumer, and consumers' first observed ZIP code, *First Zip5*), and adds last observed ZIP code (*Last Zip5*), longest observed ZIP code (*Longest Zip5*), and the number of unique ZIP codes observed (*Number Zip5*) as a measure of consumer mobility. The units in Panel A columns 1 to 4 are credit score points, and the units in Panel A columns 5 to 8 are percentage points. Panel B uses the same specification. The outcomes in Panel B are: the number of credit cards at age 30 (columns 1 and 2) and at age 40 (columns 3 and 4); the dollar value of credit card limits at age 30 (columns 5 and 6) and at age 40 (columns 7 and 8). The units in columns 1 to 4 are number of credit cards, and in columns 5 to 8 are dollars. Standard errors are clustered by birth year $\times$ the first two digits of a consumer's first observed ZIP code. $*p < .05$; $**p < .01$; $***p < .005$.

Panel A: Credit score outcomes

	Credit Score by...				Probability of Prime or Higher Score by...			
	Age 30	Age 30	Age 40	Age 40	Age 30	Age 30	Age 40	Age 40
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Immigrant	22.21*** (0.36)	14.48*** (0.54)	22.99*** (0.34)	12.59*** (0.46)	3.78*** (0.15)	6.05*** (0.22)	10.52*** (0.14)	6.33*** (0.20)
SSN Age		1.91*** (0.10)		2.43*** (0.09)		-0.53*** (0.05)		0.98*** (0.04)
F.E. Birth Year	X	X	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X	X	X
R^2	0.202	0.202	0.191	0.191	0.155	0.155	0.149	0.149
N	6,001,282	6,001,282	6,246,734	6,246,734	6,293,372	6,293,372	6,293,372	6,293,372
Mean, SSN Age <21	624.21	624.21	658.13	658.13	37.63	37.63	49.33	49.33

Panel B: Credit card outcomes

	Number of Credit Cards At...				Credit Card Limits At...			
	Age 30	Age 30	Age 40	Age 40	Age 30	Age 30	Age 40	Age 40
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Immigrant	-0.13*** (0.02)	1.12*** (0.02)	0.57*** (0.02)	0.15*** (0.02)	-2,748.6*** (98.0)	3,027.2*** (90.8)	2,296.3*** (140.5)	1,539.3*** (167.5)
SSN Age		-0.29*** (0.00)		0.09*** (0.00)		-1,345.2*** (21.0)		168.5*** (32.0)
F.E. Birth Year	X	X	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X	X	X
R^2	0.126	0.128	0.096	0.096	0.141	0.143	0.182	0.182
N	6,293,372	6,293,372	4,914,884	4,914,884	6,293,372	6,293,372	4,914,884	4,914,884
Mean, SSN Age <21	3.35	3.35	3.92	3.92	\$11,491.7	\$11,491.7	\$22,112.0	\$22,112.0

Table A4: Credit outcomes by age 37, conditional on age 30 credit score

Odd-numbered columns in both panels are estimates from Equation 1; even-numbered columns are estimates from Equation 2. The outcome in Panel A is whether a consumer has any delinquency, measured by 90 or more days past due, by age 37, in percentage points. Each column of Panel A shows results for a different credit score group at age 30. Panel A, columns 1 and 2 show results for all credit scores. Panel A, columns 3 through 12 show results for the subsets of consumers with subprime (≤ 600 , the highest credit risk segment), nearprime (601-660), prime (661-720), prime plus (721-780), and superprime (781+, the lowest risk segment) credit scores. The outcome in Panel B columns 1 and 2 is the probability of having ever held an auto loan by age 37. In columns 3 and 4 it is the probability of having ever held a mortgage by age 37; in columns 5 and 6 it is the probability of having ever held a credit card by age 37. The units in columns 1 to 6 are percentage points. The outcome in columns 7 and 8 is the number of credit cards held. The outcome in columns 9 and 10 is the value of credit card limits, measured in dollars. Standard errors are clustered by birth year $\times$ the first two digits of a consumer's first observed ZIP code. $*p < .05$; $**p < .01$; $***p < .005$.

Panel A: Any delinquency (90+ days past due) by age 37, split by credit score at age 30

Dep Var: Any delinquency by age 37	<u>All</u>		<u>Subprime</u>		<u>Nearprime</u>		<u>Prime</u>		<u>Prime Plus</u>		<u>Superprime</u>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Immigrant	-1.50*** (0.10)	-1.23*** (0.16)	-2.64*** (0.21)	-3.31*** (0.36)	-3.51*** (0.23)	-2.65*** (0.41)	-1.44*** (0.16)	0.18 (0.31)	0.13 (0.09)	1.08*** (0.17)	0.29* (0.11)	0.42 (0.23)
SSN Age		-0.07* (0.03)		0.18* (0.08)		-0.20* (0.08)		-0.36*** (0.06)		-0.22*** (0.03)		-0.04 (0.06)
F.E. Birth Year	X	X	X	X	X	X	X	X	X	X	X	X
F.E. Age 30 Zip5	X	X	X	X	X	X	X	X	X	X	X	X
F.E. Age 30 Credit Score	X	X	X	X	X	X	X	X	X	X	X	X
R^2	0.185	0.185	0.032	0.032	0.028	0.028	0.027	0.027	0.023	0.023	0.016	0.016
N	5,802,755	5,802,755	2,534,973	2,534,973	900,676	900,676	876,362	876,362	1,003,211	1,003,211	465,391	465,391
Mean, SSN Age <21	27.18	27.18	45.45	45.45	26.17	26.17	13.11	13.11	4.52	4.52	1.63	1.63

Panel B: Credit outcomes by age 37

Dep Var: Outcomes at Age 37...	<u>Any Auto Loan</u>		<u>Any Mortgage</u>		<u>Any Credit Card</u>		<u>Number of Credit Cards</u>		<u>Credit Card Limits</u>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Immigrant	-15.01*** (0.20)	-9.50*** (0.24)	-12.05*** (0.20)	-2.71*** (0.25)	-1.55*** (0.08)	-1.85*** (0.12)	0.36*** (0.01)	0.17*** (0.02)	-937.9*** (147.1)	732.3*** (179.4)
SSN Age		-1.35*** (0.04)		-2.29*** (0.05)		0.07*** (0.02)		0.05*** (0.00)		-409.3*** (35.1)
F.E. Birth Year	X	X	X	X	X	X	X	X	X	X
F.E. Age 30 Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Age 30 Credit Score	X	X	X	X	X	X	X	X	X	X
R^2	0.108	0.108	0.267	0.268	0.110	0.110	0.141	0.141	0.305	0.305
N	5,802,755	5,802,755	5,802,755	5,802,755	5,802,755	5,802,755	5,802,755	5802755	5,802,755	5,802,755
Mean, SSN Age <21	75.1	75.1	44.54	44.54	92.06	92.06	3.9	3.9	\$20,431.9	\$20,431.9

Table A5: Cumulative years of delayed access to credit from paired cohorts

This table presents estimates of the cumulative years of delayed access to credit due to immigration occurring a single year later, obtained from the paired cohort OLS regression strategy. Each column shows the results of a separate regression on a different outcome, and each row shows estimates for a different event time coefficient for each cumulative outcome. The row “C” indicates the event time year in which the control group (which immigrated 1 year before the treatment group, i.e., $C \equiv T - 1$) is assigned an SSN while “T” is the event time year of the focal (or treatment) cohort’s immigration. The specification includes all year lags from event time C through to $T + 13$, thirteen years after the treatment cohort’s immigration. The omitted category is $T - 2$. 95% confidence intervals are shown in $[\cdot]$, which are calculated using wild cluster bootstrapping by birth year using Webb weights and 10,000 iterations. N Consumers is the number of consumers across all pairs, a consumer is counted twice if they are in a focal cohort for one pair and in a control cohort for a different pair. $*p < .05$; $**p < .01$; $***p < .005$.

Dep. Var: Any ...	<u>Credit Score</u> (1)	<u>Credit Product</u> (2)	<u>Auto Loan</u> (3)	<u>Mortgage</u> (4)	<u>Credit Card</u> (5)
C	-0.076*** [-0.080, -0.072]	-0.243*** [-0.251, -0.235]	-0.030*** [-0.034, -0.026]	-0.005*** [-0.006, -0.003]	-0.227*** [-0.235, -0.218]
T	-0.390*** [-0.403, -0.377]	-0.476*** [-0.490, -0.463]	-0.090*** [-0.100, -0.080]	-0.021*** [-0.027, -0.015]	-0.448*** [-0.463, -0.434]
$T + 5$	-0.716*** [-0.745, -0.686]	-0.684*** [-0.717, -0.653]	-0.327*** [-0.347, -0.308]	-0.160*** [-0.195, -0.126]	-0.656*** [-0.689, -0.625]
$T + 10$	-0.778*** [-0.813, -0.741]	-0.725*** [-0.762, -0.689]	-0.443*** [-0.473, -0.412]	-0.283*** [-0.338, -0.228]	-0.692*** [-0.729, -0.656]
N Paired Cohorts	68	68	68	68	68
N Consumers	463,542	463,542	463,542	463,542	463,542
N	7,416,672	7,416,672	7,416,672	7,416,672	7,416,672

Table A6: Consumers still observed in 2024

Odd-numbered columns in both panels of this table are estimates from the cross-sectional OLS regression specified in Equation 1 that includes an indicator for immigration status (*Immigrant* is 1 if a consumer's SSN was assigned at age 21 or older). Even-numbered columns in both panels of this table are estimates from the OLS regression specified in Equation 2 that contains both the indicator for immigration status, and also *SSN Age*, the consumer's age at SSN assignment (pooling consumers with SSN Age 20 or younger into one group). All six regressions include fixed effects for: the birth year of the consumer, their first observed ZIP code (First Zip5), last observed ZIP code (Last Zip5), longest ZIP code (Longest Zip5) as a proxy for their most permanent location, and the number of unique ZIP codes (Number Zip5) as a proxy for their mobility. The outcome in columns 1 and 2 of Panel A is the probability of a consumer having a non-missing credit report in 2024. The outcome in columns 3 and 4 of Panel A is the probability of a consumer having a non-missing credit score in 2024. The outcome in columns 5 and 6 of Panel A is the probability of a consumer having a non-missing credit tradeline in 2024. Panel B replicates Panel B of Table 4 restricting the data to only the consumers with a non-missing credit score in 2024, and also including additional geographic fixed effects. The units are percentage points. Standard errors are clustered by birth year $\times$ the first two digits of a consumer's first observed ZIP code. * $p < .05$; ** $p < .01$; *** $p < .005$.

Panel A: Probability of consumer still being observed in 2024

Dep Var: In 2024, has...	<u>Any Credit Report</u>		<u>Any Credit Score</u>		<u>Any Credit Tradeline</u>	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	-2.10*** (0.10)	-4.38*** (0.16)	-6.25*** (0.18)	-11.19*** (0.24)	-7.39*** (0.21)	-11.83*** (0.25)
SSN Age		0.53*** (0.02)		1.15*** (0.04)		1.04*** (0.04)
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X
R^2	0.045	0.045	0.112	0.113	0.147	0.147
N	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372
Mean, SSN Age <21	95.92	95.92	89.94	89.94	84.34	84.34

Panel B: Credit access by age 37, for consumers still observed in 2024

Dep Var: By Age 37, has...	<u>Auto Loan</u>		<u>Mortgage</u>		<u>Credit Card</u>	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	-10.75*** (0.17)	-3.98*** (0.22)	-4.48*** (0.18)	4.88*** (0.23)	-0.49*** (0.07)	0.14 (0.09)
SSN Age		-1.56*** (0.04)		-2.15*** (0.05)		-0.14*** (0.02)
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X
R^2	0.11	0.111	0.138	0.138	0.061	0.061
N	5,624,238	5,624,238	5,624,238	5,624,238	5,624,238	5,624,238
Mean, SSN Age <21	79.38	79.38	48.03	48.03	95.38	95.38

Table A7: American Community Survey: income, with additional fixed effects for employment status and occupation

This table uses public data from the 5-year American Community Survey (ACS) waves 2009, 2013, 2018, and 2023, keeping consumers with birth years 1975 to 1987, non-zero income, and immigrants who immigrated between ages 21 and 29 along with non-immigrants. Each column shows results of a separate cross-sectional regression. All columns include an indicator for immigration status (*Immigrant* is an indicator for whether the consumer's age of immigration was at age 21 or older). Even-numbered columns also include *Immigration Age*, the consumer's age at immigration (pooling consumers with an age of immigration of 20 or younger into one group) to the specification. The outcome is Log (Income), the log of household income that is winsorized at 99.9th percentile. The baseline mean is the household income in dollars for non-immigrants. All regressions also include fixed effects for the consumer's birth year, geographic area (Consistent Public Use Microdata Areas), sex, education levels, race, hispanic, survey year, employment status, and occupation. Units are log points and approximate percentage differences. Heteroskedasticity-robust standard errors. ⁺ $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .005$.

	Log (Income)							
	2005 - 2009		2009 - 2013		2014 - 2018		2019 - 2023	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Immigrant	-11.23*** (0.36)	-7.57*** (0.65)	-10.98*** (0.30)	-6.50*** (0.54)	-9.15*** (0.27)	-5.43*** (0.51)	-4.41*** (0.33)	-1.27* (0.61)
Immigration Age		-0.93*** (0.14)		-1.05*** (0.11)		-0.79*** (0.09)		-0.66*** (0.11)
F.E. Birth Year	X	X	X	X	X	X	X	X
F.E. Microdata Area	X	X	X	X	X	X	X	X
F.E. Sex	X	X	X	X	X	X	X	X
F.E. Education	X	X	X	X	X	X	X	X
F.E. Race	X	X	X	X	X	X	X	X
F.E. Hispanic	X	X	X	X	X	X	X	X
F.E. Survey Year	X	X	X	X	X	X	X	X
F.E. Employment	X	X	X	X	X	X	X	X
F.E. Occupation	X	X	X	X	X	X	X	X
<i>N</i>	2,079,500	2,079,500	2,162,012	2,162,012	2,231,275	2,231,275	2,200,018	2,200,018
<i>R</i> ²	0.199	0.199	0.248	0.248	0.299	0.299	0.299	0.299
Mean Income, Immigration Age <21	\$71,088	\$71,088	\$78,339	\$78,339	\$98,660	\$98,660	\$134,099	\$134,099

Table A8: American Community Survey: mortgage use, homeownership, and car ownership

This table uses public data from the 5-year American Community Survey (ACS) waves 2005-2009, 2009-2013, 2014-2018, and 2019-2023, keeping consumers with birth years 1975 to 1987, non-zero income, and immigrants who immigrated between ages 21 and 29 along with non-immigrants. Each column in each panel shows results of a separate cross-sectional regression. The outcome in Panel A is whether a consumer has any mortgage, the outcome in Panel B is whether a consumer is a homeowner (with or without a mortgage), and the outcome in Panel C is whether the consumer has any car. All columns include an indicator for immigration status (*Immigrant* is an indicator for whether the consumer's age of immigration was at age 21 or older) and *Immigration Age*, the consumer's age at immigration (pooling consumers with an age of immigration of 20 or younger into one group) to the specification. All regressions also include fixed effects for the consumer's birth year, geographic area (Consistent Public Use Microdata Areas), sex, education levels, race, hispanic, and survey year. All columns except 1, 4, 7, and 10 (in all panels) include a control for Log (Income), the log of household income that is winsorized at 99.9th percentile and is scaled to mean zero and standard deviation one. In all panels, columns 3, 6, 9, and 12 also include additional fixed effects for employment status, and occupation. Units are percentage points. Heteroskedasticity-robust standard errors. ⁺ $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .005$.

	2005 - 2009			2009 - 2013			2014 - 2018			2019 - 2023		
A. Dep Var: Any Mortgage	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Immigrant	-9.90*** (0.31)	-7.57*** (0.30)	-6.90*** (0.30)	-8.09*** (0.28)	-6.21*** (0.27)	-5.38*** (0.27)	-3.56*** (0.29)	-2.09*** (0.28)	-1.23*** (0.28)	-1.86*** (0.35)	-1.40*** (0.34)	-0.65* (0.34)
Immigration Age	-1.98*** (0.06)	-1.69*** (0.06)	-1.73*** (0.06)	-1.63*** (0.05)	-1.32*** (0.05)	-1.34*** (0.05)	-1.36*** (0.05)	-1.14*** (0.05)	-1.13*** (0.05)	-0.76*** (0.06)	-0.60*** (0.06)	-0.56*** (0.06)
Log (Income)		17.92*** (0.05)	18.13*** (0.05)		16.43*** (0.05)	16.41*** (0.05)		15.23*** (0.05)	14.88*** (0.05)		13.88*** (0.06)	13.12*** (0.06)
R ²	0.100	0.213	0.220	0.116	0.205	0.212	0.135	0.206	0.213	0.131	0.192	0.199
Mean, Immigration Age <21	44.22	44.22	44.22	44.65	44.65	44.65	48.12	48.12	48.12	54.25	54.25	54.25
B. Dep Var: Homeowner	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Immigrant	-13.08*** (0.32)	-10.56*** (0.32)	-10.19*** (0.32)	-10.65*** (0.30)	-8.67*** (0.29)	-8.04*** (0.29)	-4.82*** (0.30)	-3.31*** (0.29)	-2.50*** (0.29)	-0.83** (0.35)	-0.37 (0.34)	0.35 (0.34)
Immigration Age	-2.02*** (0.07)	-1.70*** (0.07)	-1.79*** (0.06)	-1.75*** (0.06)	-1.42*** (0.06)	-1.50*** (0.06)	-1.56*** (0.05)	-1.34*** (0.05)	-1.38*** (0.05)	-1.10*** (0.06)	-0.93*** (0.06)	-0.94*** (0.06)
Log (Income)		19.37*** (0.05)	19.91*** (0.05)		17.31*** (0.05)	17.78*** (0.05)		15.62*** (0.05)	15.86*** (0.05)		13.80*** (0.06)	13.71*** (0.06)
R ²	0.114	0.244	0.255	0.128	0.225	0.235	0.152	0.229	0.237	0.155	0.224	0.233
Mean, Immigration Age <21	52.40	52.40	52.40	53.25	53.25	53.25	59.26	59.26	59.26	68.54	68.54	68.54
C. Dep Var: Any Car	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Immigrant	-2.37*** (0.25)	-1.72*** (0.24)	-1.34*** (0.24)	-0.88*** (0.20)	-0.34* (0.19)	0.00 (0.19)	1.11*** (0.16)	1.53*** (0.16)	1.69*** (0.16)	1.18*** (0.17)	1.30*** (0.17)	1.39*** (0.17)
Immigration Age	-0.65*** (0.05)	-0.56*** (0.05)	-0.54*** (0.05)	-0.57*** (0.04)	-0.48*** (0.04)	-0.45*** (0.04)	-0.41*** (0.03)	-0.35*** (0.03)	-0.33*** (0.03)	-0.16*** (0.03)	-0.11*** (0.03)	-0.10*** (0.03)
Log (Income)		4.99*** (0.03)	4.85*** (0.03)		4.80*** (0.03)	4.61*** (0.03)		4.36*** (0.03)	4.15*** (0.03)		3.67*** (0.03)	3.48*** (0.03)
R ²	0.200	0.234	0.236	0.211	0.240	0.243	0.187	0.215	0.219	0.153	0.177	0.181
Mean, Immigration Age <21	93.51	93.51	93.51	93.66	93.66	93.66	94.69	94.69	94.69	95.40	95.40	95.40
F.E. Demographics	X	X	X	X	X	X	X	X	X	X	X	X
F.E. Employment & Occupation			X			X			X			X
N	2,079,500	2,079,500	2,079,500	2,162,012	2,162,012	2,162,012	2,231,275	2,231,275	2,231,275	2,200,018	2,200,018	2,200,018

Table A9: Survey of Consumer Finances (2022): debt attitudes

This table uses public data from the 2022 Survey of Consumer Finances (SCF), keeping birth years 1972 to 2000 such that respondents are aged between 22 and 49. Odd-numbered columns are estimates from the cross-sectional OLS regression that includes an indicator for immigration status (*Immigrant* is an indicator for whether the consumer's age of immigration was at age 21 or older). Even-numbered columns add *Immigration Age*, the consumer's age at immigration (pooling consumers with an age of immigration of 20 or younger into one group) to the specification. See Appendix A.2 for details of the survey variables used. The units are percentage points, except for columns 7 and 8 where the estimates are multiplied by 100 to be on a similar scale to percentage points (this outcome takes values -1, 0, and +1). All regressions include a control for normal income (winsorized at the 99th percentile), and fixed effects for the birth year of the consumer and fixed effects for other demographic control variables: male, race, Hispanic, spouse, marital status, household size, number of children, and education levels. Columns 3 to 6 also include a control for the binary auto loan debt attitude variable, where a value of 1 is considering borrowing for auto is all right. Heteroskedasticity-robust standard errors. Missing data in the SCF are imputed five times using a multiple imputation technique, storing data in five "implicates". We run a separate regression on each of the five implicates and follow the SCF's recommended multiply-imputed variance estimation technique for combining standard errors. ⁺ $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .005$.

Dep Var:	<u>Auto Debt Attitude</u>		<u>Any Auto Debt</u>		<u>Any Auto Debt</u>		<u>General Debt Attitude</u>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Immigrant	-15.41*** (4.58)	-23.02*** (7.25)	-4.27 (5.08)	-11.14 (8.27)	-0.21 (5.01)	-5.10 (7.87)	-8.96 (7.92)	-15.94 (13.71)
Immigration Age		0.98 (0.62)		0.88 (0.76)		0.63 (0.69)		0.89 (1.19)
Auto Debt Attitude					26.34*** (3.23)	26.24*** (3.24)		
Demographic Controls	X	X	X	X	X	X	X	X
R^2	0.049	0.050	0.063	0.063	0.106	0.106	0.026	0.026
N	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813
Mean, Immigration Age <21	79.58	79.58	47.14	47.14	47.14	47.14	3.80	3.80

Table A10: Survey of Consumer Finances (2022): debt attitudes and credit demand

This table uses public data from the 2022 Survey of Consumer Finances (SCF), keeping birth years 1972 to 2000 such that respondents are aged between 22 and 49. Odd-numbered columns in both panels are estimates from the cross-sectional OLS regression that includes an indicator for immigration status (*Immigrant* is an indicator for whether the consumer's age of immigration was at age 21 or older). Even-numbered columns add *Immigration Age*, the consumer's age at immigration (pooling consumers with an age of immigration of 20 or younger into one group) to the specification. See Appendix A.2 for details of the survey variables used. The outcomes in Panel A are a variety of measures of attitudes to debt, as described more in Appendix A.2. The outcomes in Panel B are indicators for whether a consumer has applied for credit in the last twelve months: The outcome in columns 1-2 is a binary variable for whether a consumer has applied for any credit in the past twelve months (constructed from variables listed below, and also variable x438 whether applied for student loan). Columns 3-4 (constructed from variable x437) is whether applied for auto loan. Columns 5-6 (constructed from variable x435) is whether applied for mortgage. Columns 7-8 is whether applied for credit card or respond to a pre-approved credit card (constructed from variable x433) or applied for a credit card limit increase (constructed from variable x434). Columns 9-10 (constructed from variable x439) is whether applied for other consumer credit. The units in both panels are percentage points. All regressions include a control for normal income (winsorized at the 99th percentile), and fixed effects for the birth year of the consumer and fixed effects for other demographic control variables: male, race, Hispanic, spouse, marital status, household size, number of children, and education levels. Heteroskedasticity-robust standard errors. Missing data in the SCF are imputed five times using a multiple imputation technique, storing data in five "implicates". We run a separate regression on each of the five implicates and follow the SCF's recommended multiply-imputed variance estimation technique for combining standard errors. R^2 is the adjusted r-squared averaged across the regressions for the five implicates. $^+p < .10$; $*p < .05$; $**p < .01$; $***p < .005$.

Panel A: Debt attitudes

Dep Var: All Right To Borrow...	<u>In General (Positive)</u>		<u>In General (Negative)</u>		<u>For Living Expenses</u>		<u>For Vacation</u>		<u>For Education</u>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Immigrant	-4.27 (4.83)	-4.29 (8.95)	4.69 (4.65)	11.65 (7.09)	-8.76 ⁺ (4.89)	-8.35 (7.66)	-5.01 (3.40)	-8.37 (5.21)	-2.80 (4.18)	9.07 (5.59)
Immigration Age		0.00 (0.80)		-0.89 (0.62)		-0.05 (0.74)		0.43 (0.56)		-1.52* (0.67)
Demographic Controls	X	X	X	X	X	X	X	X	X	X
R^2	0.016	0.016	0.043	0.044	0.032	0.032	0.020	0.020	0.040	0.045
N	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1813
Mean, Immigration Age <21	29.51	29.51	25.71	25.71	73.38	73.38	15.30	15.30	84.24	84.24

Panel B: Credit demand

Dep Var: Apply For...	<u>Any Credit</u>		<u>Auto</u>		<u>Mortgage</u>		<u>Credit Card</u>		<u>Other Credit</u>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Immigrant	2.32 (4.56)	-1.08 (7.17)	2.03 (4.37)	-6.55 (7.53)	2.11 (3.80)	5.65 (6.85)	1.63 (3.22)	2.02 (5.32)	1.84 (3.62)	-7.66 ⁺ (4.61)
Immigration Age		0.44 (0.62)		1.10 (0.72)		-0.45 (0.65)		-0.05 (0.48)		1.22* (0.55)
Demographic Controls	X	X	X	X	X	X	X	X	X	X
R^2	0.051	0.051	0.051	0.053	0.045	0.045	0.045	0.045	0.026	0.029
N	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813
Mean, Immigration Age <21	64.81	64.81	19.95	19.95	11.28	11.28	9.28	9.28	12.08	12.08